



IIT ROORKEE



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CERTIFICATION COURSE

Lecture 11: Python demo for Distribution

Dr. A. Ramesh

DEPARTMENT OF MANAGEMENT STUDIES



Agenda

- Different numerical problems are solved for the following Distribution using Python:
 - Discrete
 - Binomial
 - Poisson
 - Hyper geometric
 - Continuous
 - Uniform
 - Exponential
 - Normal

THANK YOU





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Lecture 11: Sampling and Sampling Distribution

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Lecture Objectives

After completing this lecture, you should be able to:

- Describe a simple random sample and why sampling is important
- Explain the difference between descriptive and inferential statistics
- Define the concept of a sampling distribution
- Determine the mean and standard deviation for the sampling distribution of the sample mean,

Lecture Objectives

- Describe the Central Limit Theorem and its importance
- Determine the mean and standard deviation for the sampling distribution of the sample proportion,
- Describe sampling distributions of sample variances

Descriptive vs Inferential Statistics

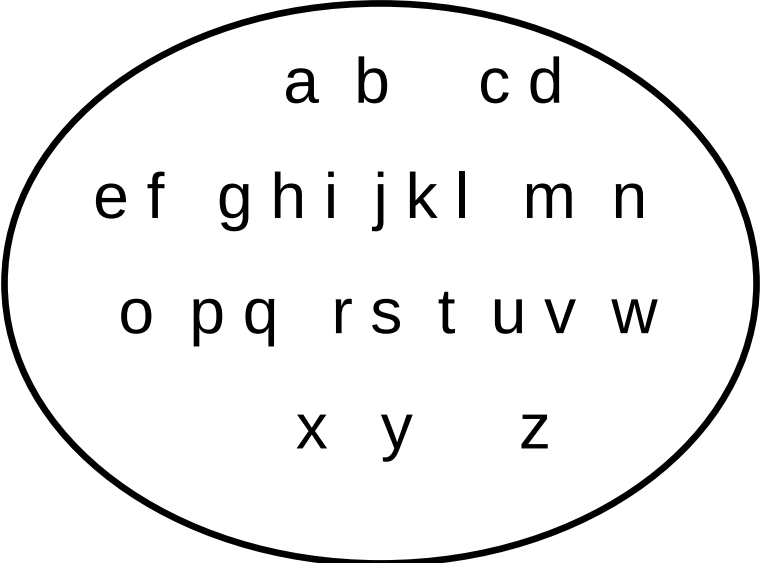
- **Descriptive statistics**
 - Collecting, presenting, and describing data
- **Inferential statistics**
 - Drawing conclusions and/or making decisions concerning a population based only on sample data

Populations and Samples

- A **Population** is the set of all items or individuals of interest
 - **Examples:**
 - All likely voters in the next election
 - All parts produced today
 - All sales receipts for November
- A **Sample** is a subset of the population
 - **Examples:**
 - 1000 voters selected at random for interview
 - A few parts selected for destructive testing
 - Random receipts selected for audit

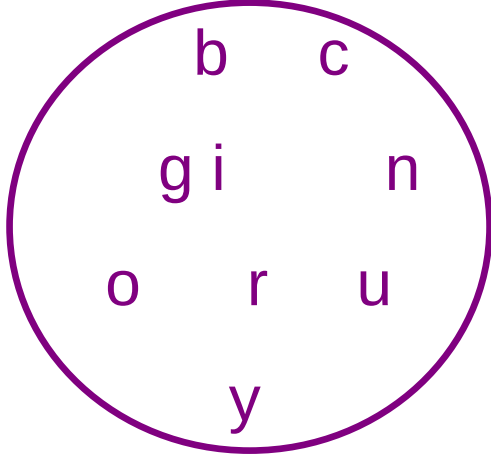
Population vs. Sample

- **Population**



a b c d
e f g h i j k l m n
o p q r s t u v w
x y z

- **Sample**



b c
g i n
o r u
y

Why Sample?

- Less time consuming than a census
- Less costly to administer than a census
- It is possible to obtain statistical results of a sufficiently high precision based on samples.
- Because the research process is sometimes destructive, the sample can save product
- If accessing the population is impossible; sampling is the only option

Reasons for Taking a Census

- Eliminate the possibility that a random sample is not representative of the population
- The person authorizing the study is uncomfortable with sample information



Random Versus Nonrandom Sampling

- **Random sampling**

- Every unit of the population has the same probability of being included in the sample.
- A chance mechanism is used in the selection process.
- Eliminates bias in the selection process
- Also known as probability sampling

- **Nonrandom Sampling**

- Every unit of the population does not have the same probability of being included in the sample.
- Open the selection bias
- Not appropriate data collection methods for most statistical methods
- Also known as non-probability sampling



Random Sampling Techniques

- Simple Random Sample
- Stratified Random Sample
 - Proportionate
 - Disproportionate
- Systematic Random Sample
- Cluster (or Area) Sampling



Simple Random Samples

- Every object in the population has an **equal chance** of being selected
- Objects are selected independently
- Samples can be obtained from a table of random numbers or computer random number generators
- A simple random sample is the ideal against which other sample methods are compared

Simple Random Sample: Numbered Population Frame

01 Andhra Pradesh
02 Himachal Pradesh
03 Gujrath
04 Maharashtra
05 Nagaland
06 Goa
07 West bengal
08 Haryana
09 Punjab
10 Delhi

11 Madhya Pradesh
12 Uttar Pradesh
13 Bihar
14 Rajasthan
15 J & K
16 Tamil Nadu
17 Karantaka
18 Kerala
19 Orissa
20 Manipur

Simple Random Sampling: Random Number Table

9 9 4 3 7	8 7 9 6 1	4 5 7 3 7	3 7 5 5 2	9 7 9 6 9	3 9 0 9 4	3 4 4 7 5	3 1 6 1 8
5 0 6 5 6	0 0 1 2 7	6 8 3 6 7	6 6 8 8 2	0 8 1 5 6	8 0 0 1 6	7 8 2 2 4	5 8 3 2 6
8 0 8 8 0	6 3 1 7 1	4 2 8 7 7	6 6 8 3 5	6 0 5 1 5	7 0 2 9 6	5 0 0 2 6	4 5 5 8 7
8 6 4 2 0	4 0 8 5 3	5 3 7 9 8	8 9 4 5 4	6 8 1 3 0	9 1 2 5 3	8 8 1 0 4	7 4 3 1 9
6 0 0 9 7	8 6 4 3 6	0 1 8 6 9	4 7 7 5 8	8 9 5 3 5	9 9 4 0 0	4 8 2 6 8	3 0 6 0 6
5 2 5 8 7	7 1 9 6 5	8 5 4 5 3	4 6 8 3 4	0 0 9 9 1	9 9 7 2 9	7 6 9 4 8	1 5 9 4 1
8 9 1 5 5	9 0 5 5 3	9 0 6 8 9	4 8 6 3 7	0 7 9 5 5	4 7 0 6 2	7 1 1 8 2	6 4 4 9 3

Simple Random Sample: Sample Members

01 Andhra Pradesh
02 Himachal Pradesh
03 Gujrath
04 Maharashtra
05 Nagaland
06 Goa
07 West bengal
08 Haryana
09 Punjab
10 Delhi

11 Madhya Pradesh
12 Uttar Pradesh
13 Bihar
14 Rajasthan
15 J & K
16 Tamil Nadu
17 Karantaka
18 Kerala
19 Orissa
20 Manipur

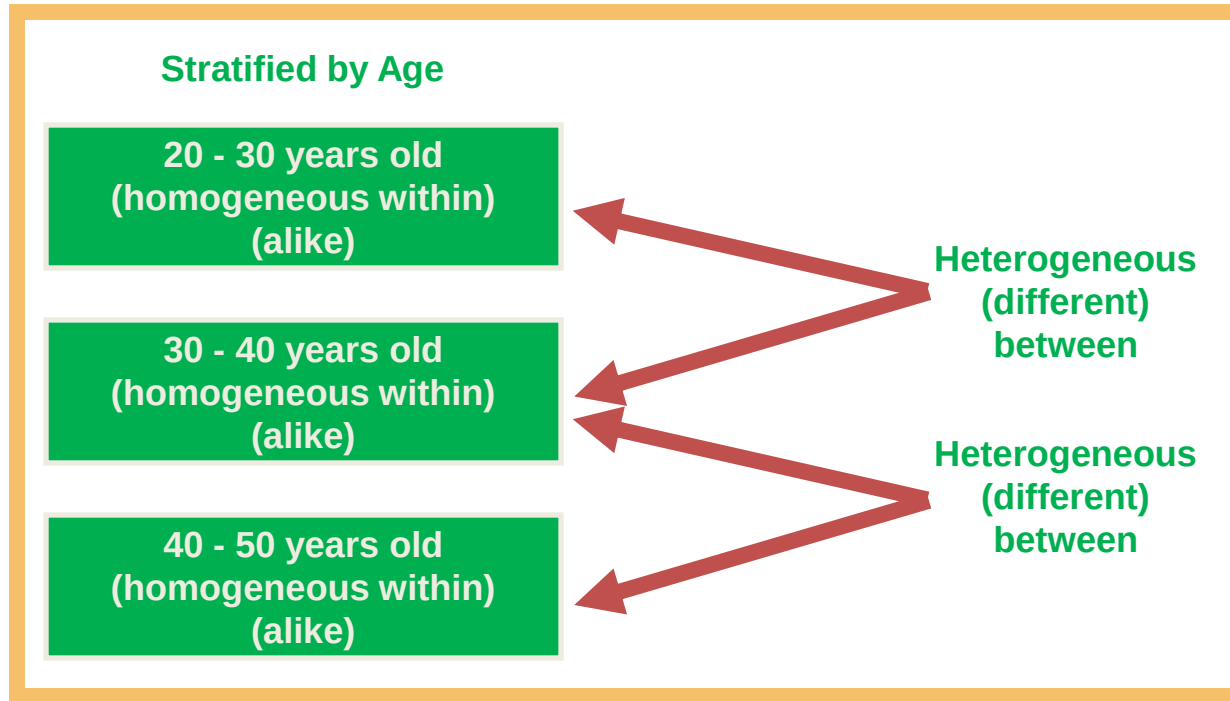
- $N = 20$
- $n = 4$

Stratified Random Sample

- Population is divided into non-overlapping subpopulations called strata
- A random sample is selected from each stratum
- Potential for reducing sampling error
- Proportionate -- the percentage of these sample taken from each stratum is proportionate to the percentage that each stratum is within the population
- Disproportionate -- proportions of the strata within the sample are different than the proportions of the strata within the population



Stratified Random Sample: Population of FM Radio Listeners



Systematic Sampling

- Convenient and relatively easy to administer
- Population elements are an ordered sequence (at least, conceptually).
- The first sample element is selected randomly from the first k population elements.
- Thereafter, sample elements are selected at a constant interval, k , from the ordered sequence frame.

$$k = \frac{N}{n} ,$$

where:

n = sample size

N = population size

k = size of selection interval

Systematic Sampling: Example

- Purchase orders for the previous fiscal year are serialized 1 to 10,000 ($N = 10,000$).
- A sample of fifty ($n = 50$) purchases orders is needed for an audit.
- $k = 10,000/50 = 200$
- First sample element randomly selected from the first 200 purchase orders. Assume the 45th purchase order was selected.
- **Subsequent sample elements: 245, 445, 645, . . .**



Cluster Sampling

- Population is divided into non-overlapping clusters or areas
- Each cluster is a miniature of the population.
- A subset of the clusters is selected randomly for the sample.
- If the number of elements in the subset of clusters is larger than the desired value of n , these clusters may be subdivided to form a new set of clusters and subjected to a random selection process.



Cluster Sampling

◆ Advantages

- More convenient for geographically dispersed populations
- Reduced travel costs to contact sample elements
- Simplified administration of the survey
- Unavailability of sampling frame prohibits using other random sampling methods

◆ Disadvantages

- Statistically less efficient when the cluster elements are similar
- Costs and problems of statistical analysis are greater than for simple random sampling

Nonrandom Sampling

- **Convenience Sampling:** Sample elements are selected for the convenience of the researcher
- **Judgment Sampling:** Sample elements are selected by the judgment of the researcher
- **Quota Sampling:** Sample elements are selected until the quota controls are satisfied
- **Snowball Sampling:** Survey subjects are selected based on referral from other survey respondents

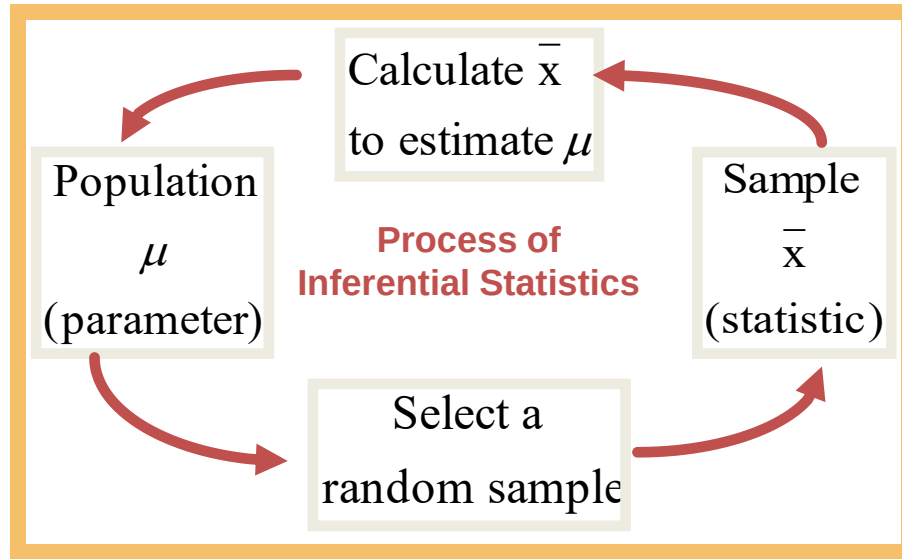


Errors

- Data from nonrandom samples are not appropriate for analysis by inferential statistical methods.
- Sampling Error occurs when the sample is not representative of the population
- Non-sampling Errors
 - Missing Data, Recording, Data Entry, and Analysis Errors
 - Poorly conceived concepts , unclear definitions, and defective questionnaires
 - Response errors occur when people so not know, will not say, or overstate in their answers

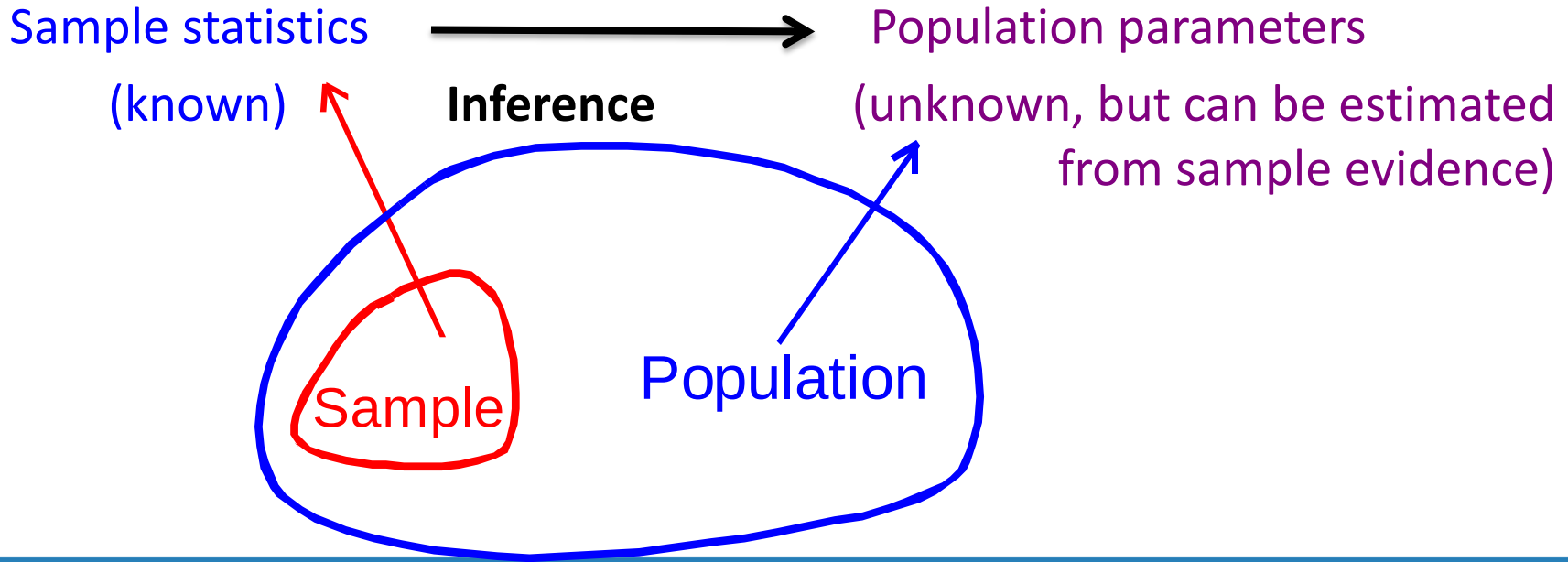
Sampling Distribution of $\bar{x}$

Proper analysis and interpretation of a sample statistic requires knowledge of its distribution.



Inferential Statistics

- Making statements about a population by examining sample results



Inferential Statistics

Drawing conclusions and/or making decisions concerning a **population** based on **sample** results.

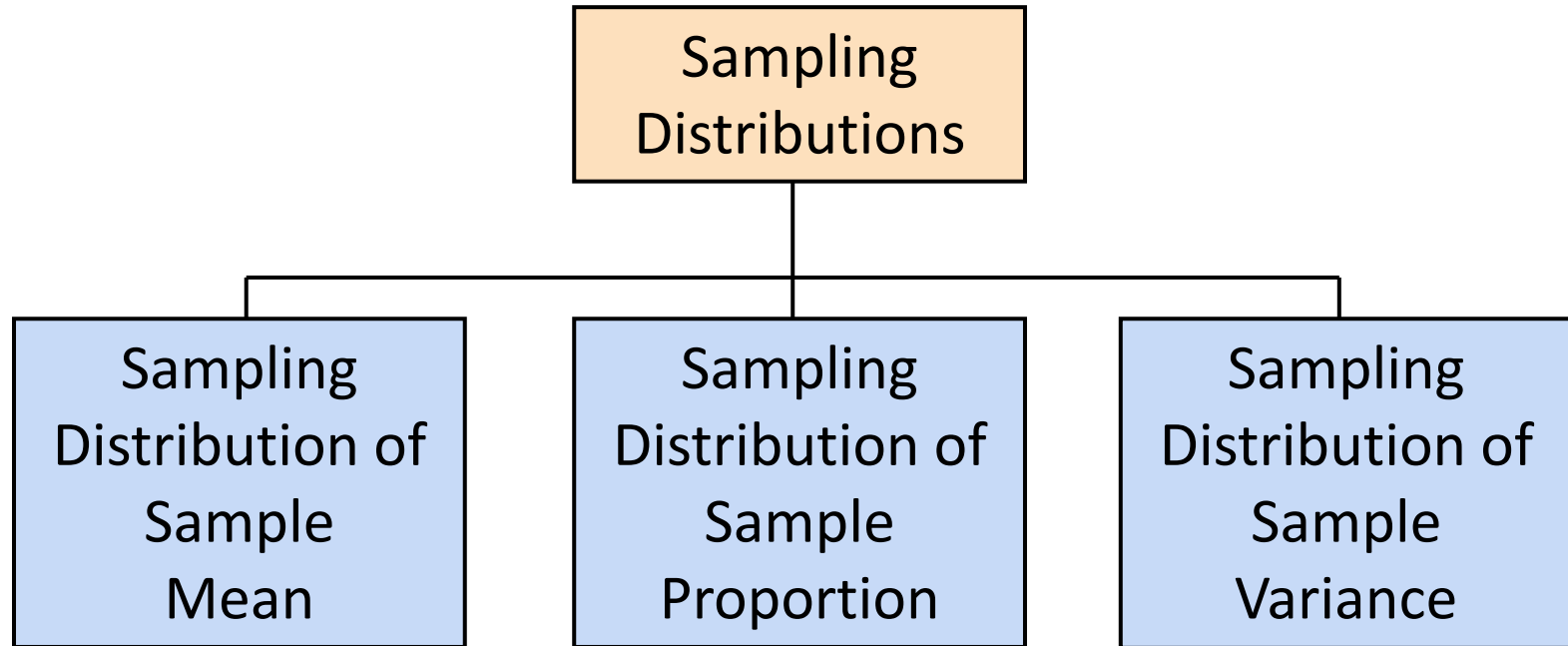
- **Estimation**
 - e.g., Estimate the population mean weight using the sample mean weight
- **Hypothesis Testing**
 - e.g., Use sample evidence to test the claim that the population mean weight is 120 pounds



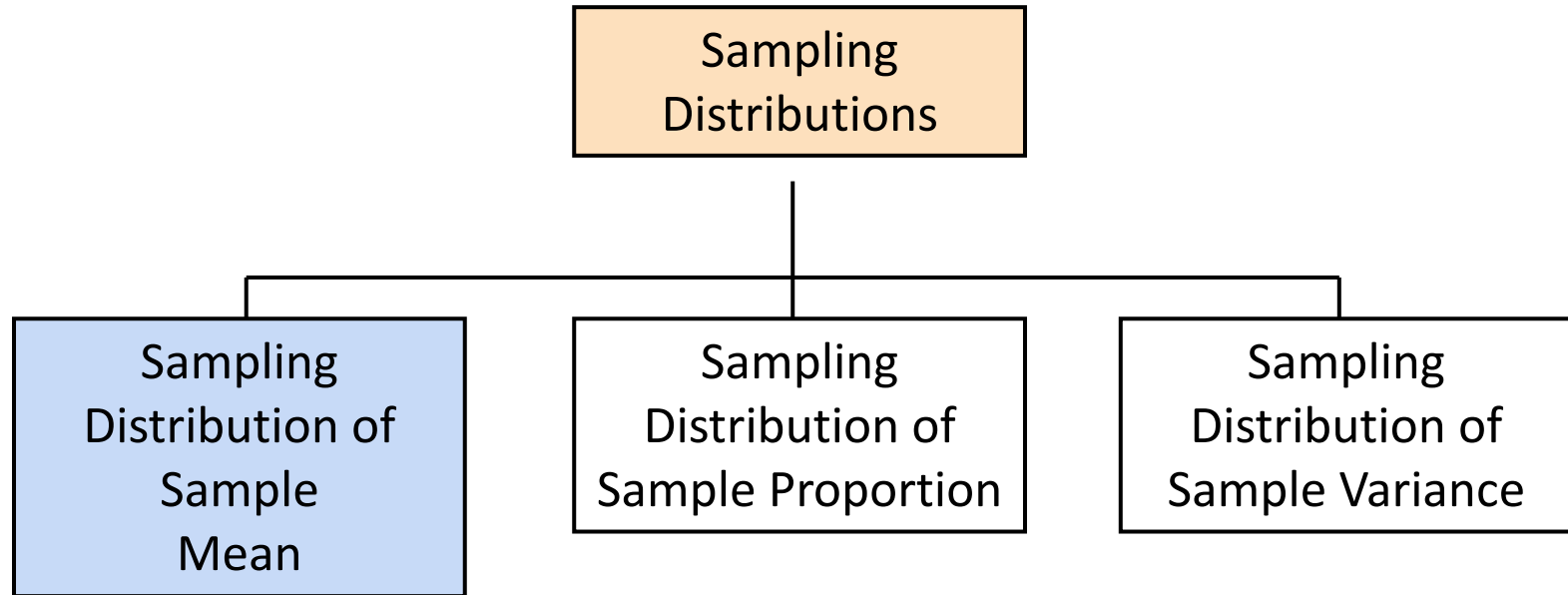
Sampling Distributions

- A **sampling distribution** is a distribution of all of the possible values of a statistic for a given size sample selected from a population

Types of sampling distributions



Sampling Distributions of Sample Means



Developing a Sampling Distribution

- Assume there is a population ...
- Population size $N=4$
- Random variable, X , is age of individuals
- Values of X :
18, 20, 22, 24 (years)

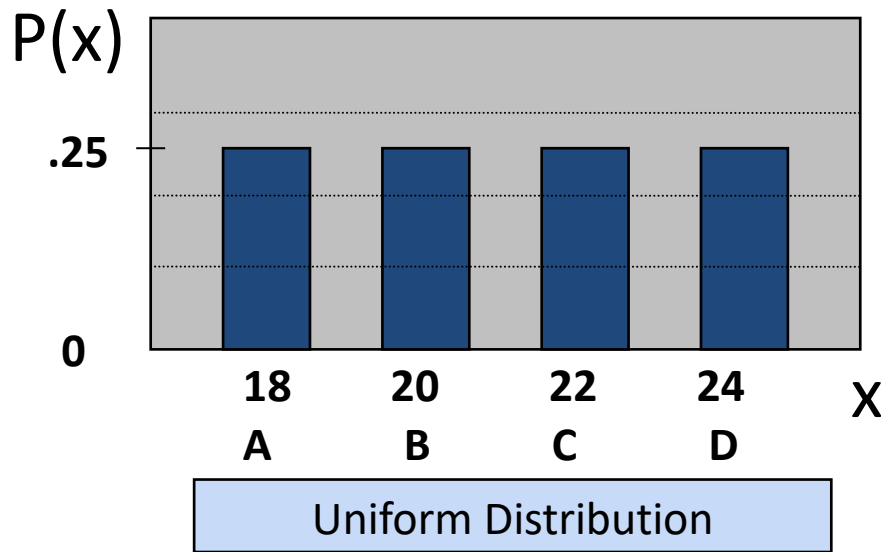


Developing a Sampling Distribution (continued)

Summary Measures for the **Population** Distribution:

$$\begin{aligned}\mu &= \frac{\sum X_i}{N} \\ &= \frac{18 + 20 + 22 + 24}{4} = 21\end{aligned}$$

$$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{N}} = 2.236$$



Developing a Sampling Distribution *(continued)*

Now consider all possible samples of size $n = 2$

1 st	2 nd Observation			
Obs	18	20	22	24
18	18,18	18,20	18,22	18,24
20	20,18	20,20	20,22	20,24
22	22,18	22,20	22,22	22,24
24	24,18	24,20	24,22	24,24

16 possible samples
(sampling with
replacement)

16 Sample
Means

1 st	2 nd Observation			
Obs	18	20	22	24
18	18	19	20	21
20	19	20	21	22
22	20	21	22	23
24	21	22	23	24

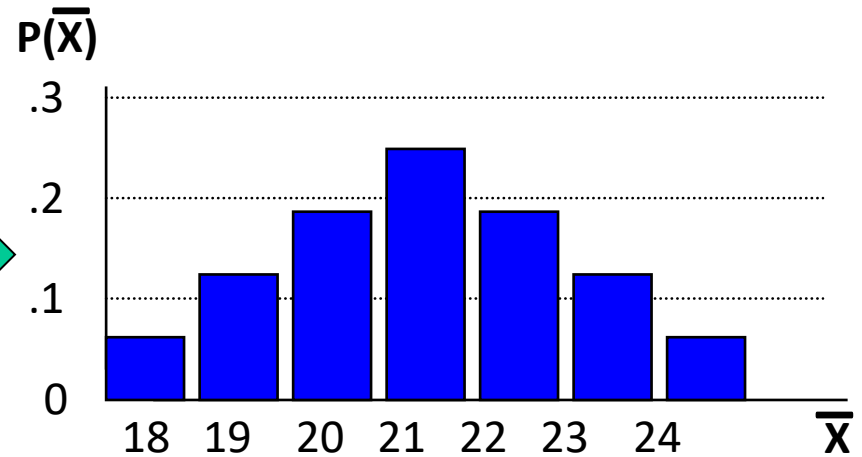
Developing a Sampling Distribution (continued)

- Sampling Distribution of All Sample Means

16 Sample Means

1st Obs	2nd Observation			
	18	20	22	24
18	18	19	20	21
20	19	20	21	22
22	20	21	22	23
24	21	22	23	24

Sample Means Distribution



(no longer uniform)

Developing a Sampling Distribution *(continued)*

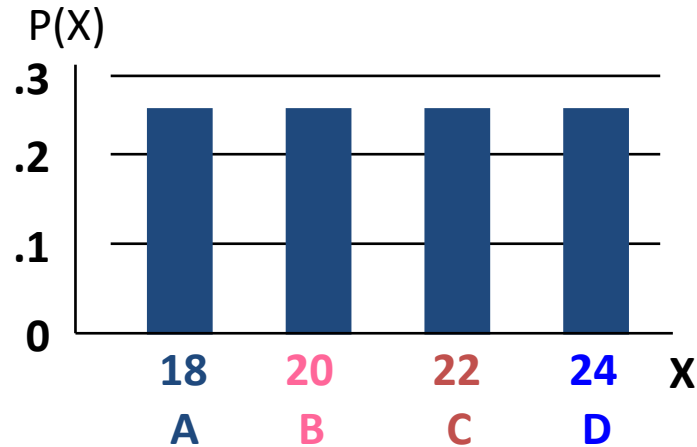
- Summary Measures of this Sampling Distribution:

$$E(\bar{X}) = \frac{\sum \bar{X}_i}{N} = \frac{18+19+21+\dots+24}{16} = 21 = \mu$$

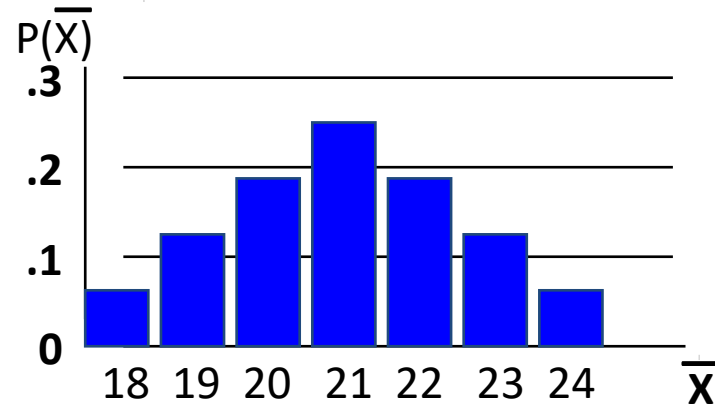
$$\begin{aligned}\sigma_{\bar{X}} &= \sqrt{\frac{\sum (\bar{X}_i - \mu)^2}{N}} \\ &= \sqrt{\frac{(18-21)^2 + (19-21)^2 + \dots + (24-21)^2}{16}} = 1.58\end{aligned}$$

Comparing the Population with its Sampling Distribution

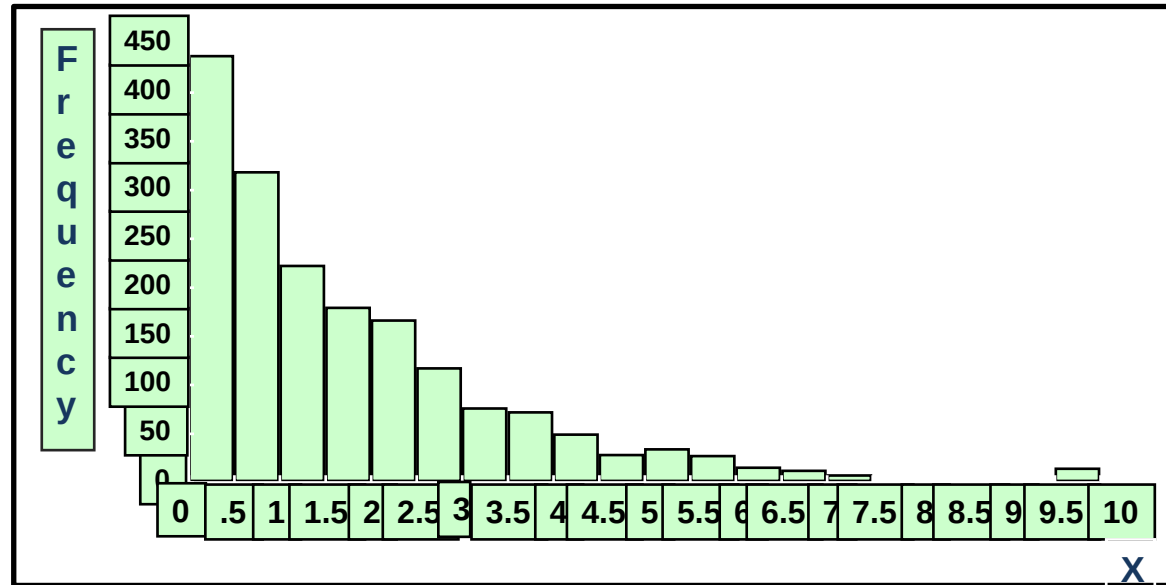
Population
 $N = 4$
 $\mu = 21$ $\sigma = 2.236$



Sample Means Distribution
 $n = 2$
 $\mu_{\bar{X}} = 21$ $\sigma_{\bar{X}} = 1.58$

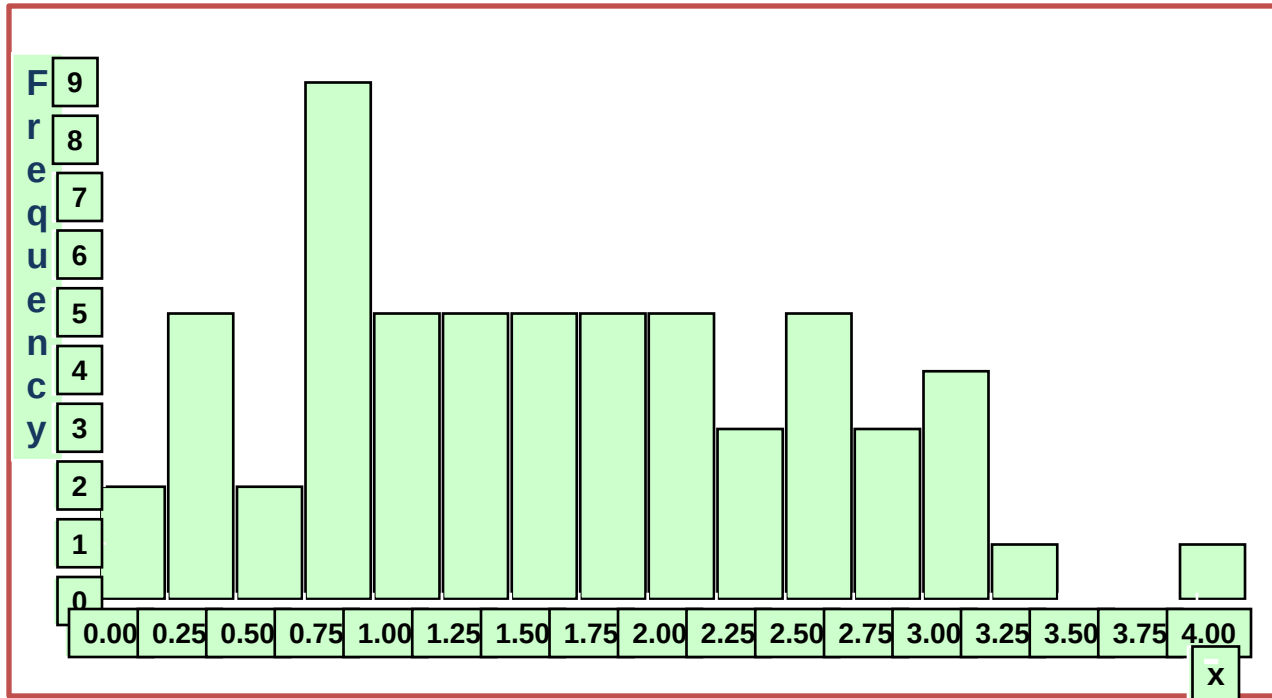


1,800 Randomly Selected Values from an Exponential Distribution

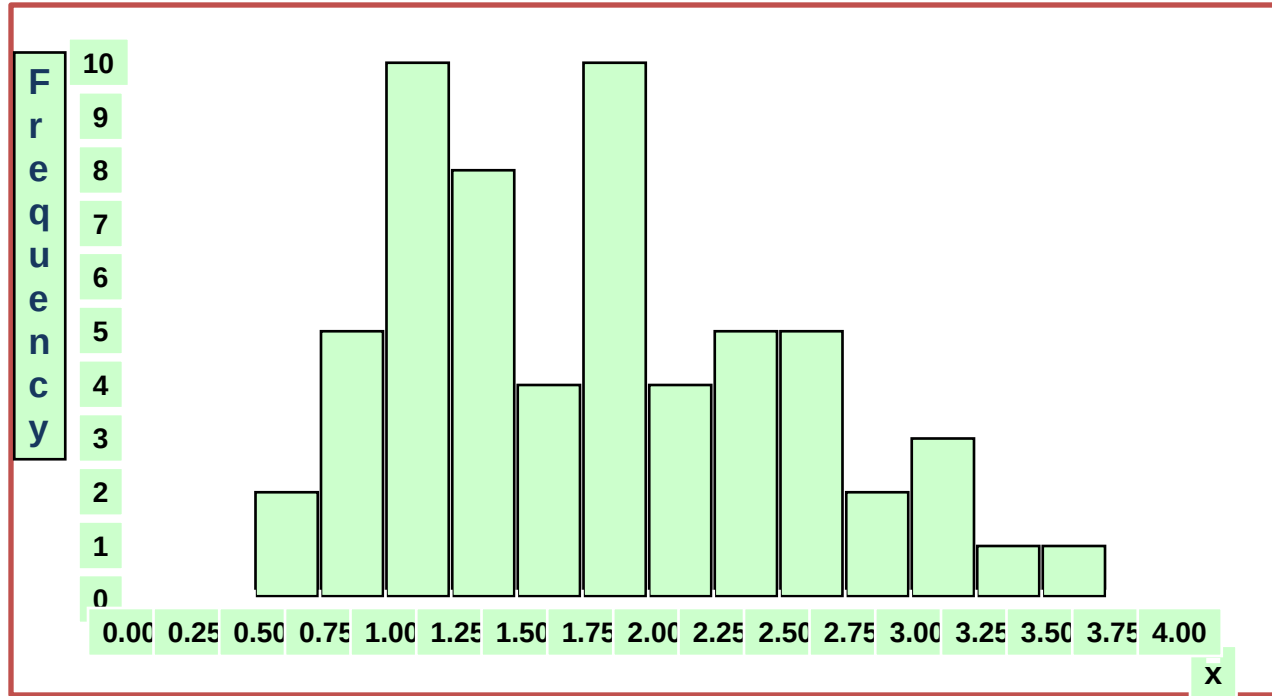


X

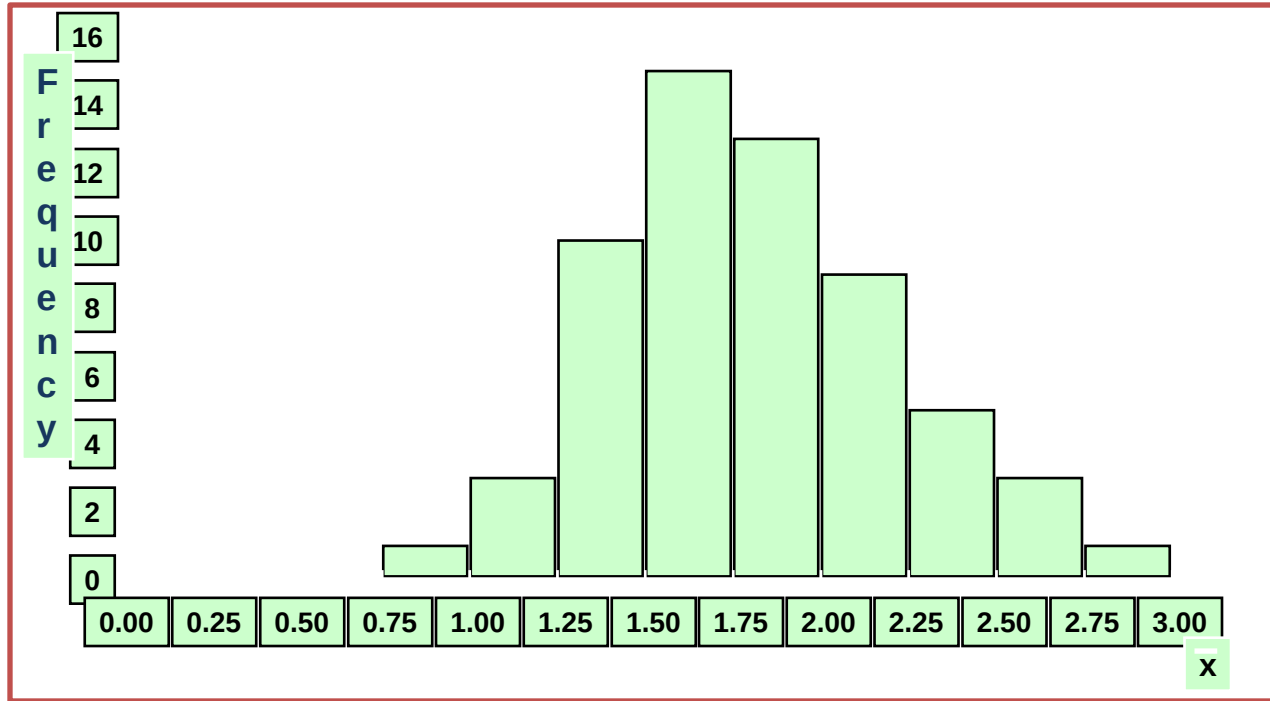
Means of 60 Samples ($n = 2$) from an Exponential Distribution



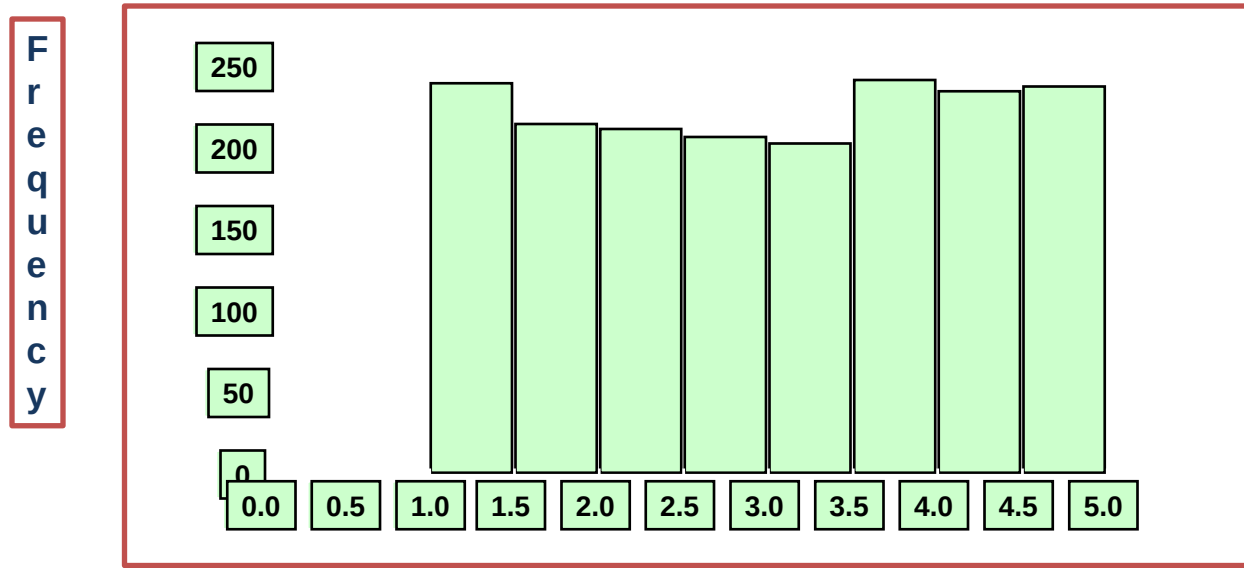
Means of 60 Samples ($n = 5$) from an Exponential Distribution



Means of 60 Samples ($n = 30$) from an Exponential Distribution

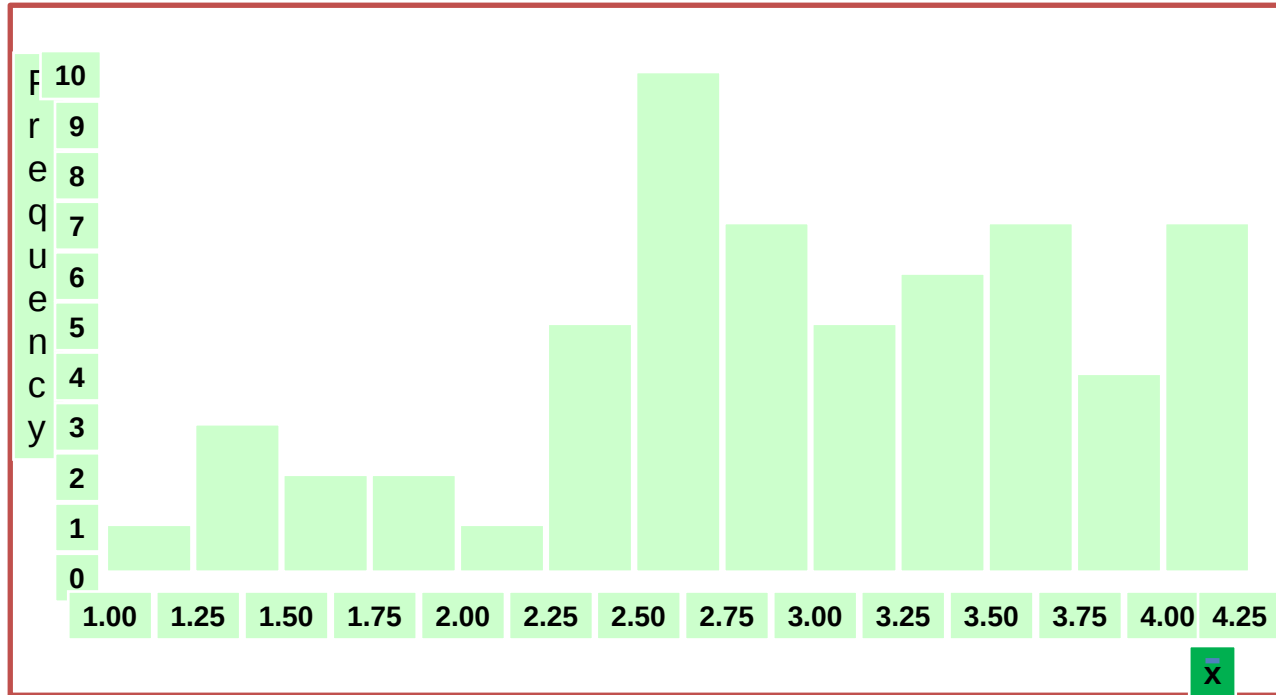


1,800 Randomly Selected Values from a Uniform Distribution



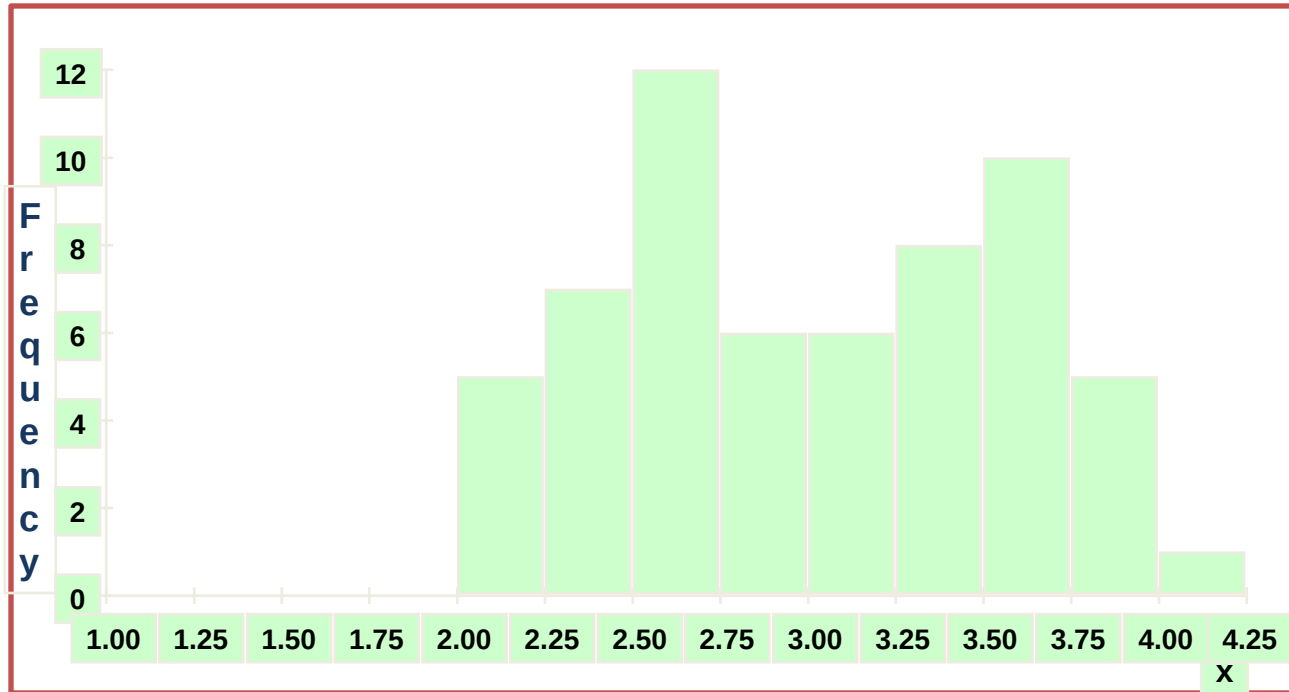
X-bar

Means of 60 Samples ($n = 2$) from a Uniform Distribution

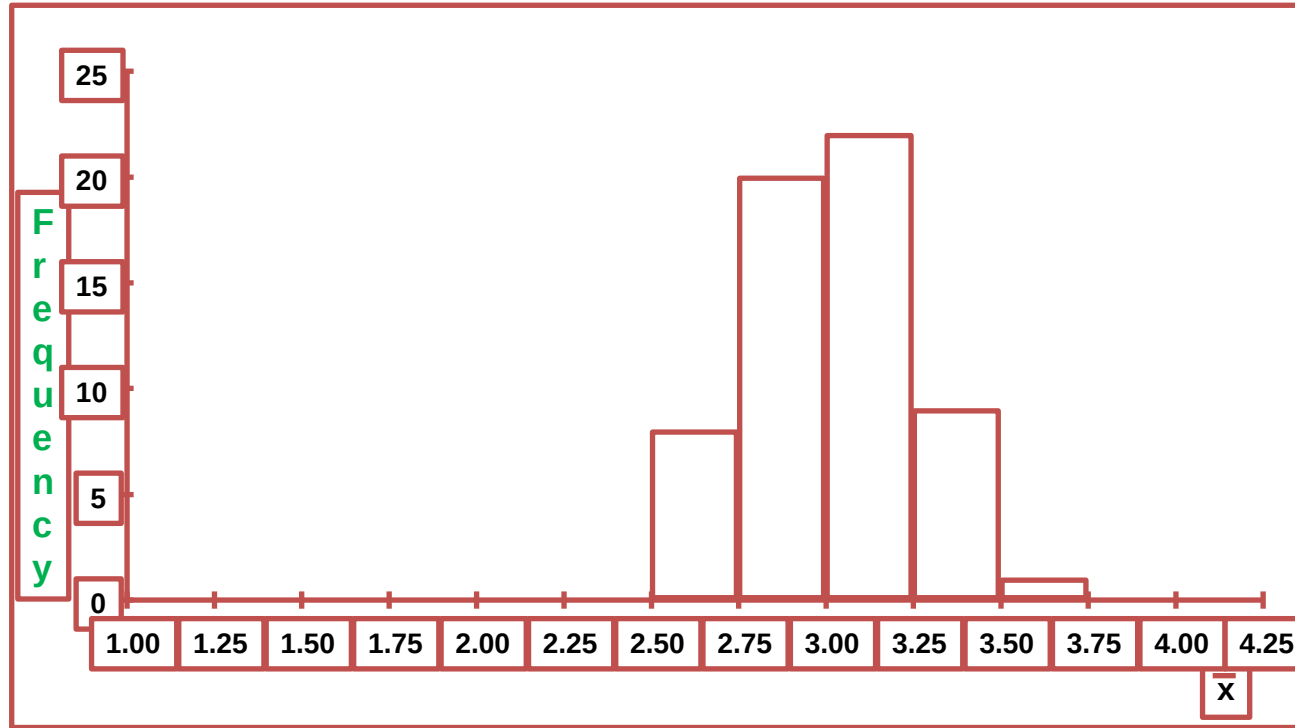


$\bar{x}$

Means of 60 Samples ($n = 5$) from a Uniform Distribution



Means of 60 Samples ($n = 30$) from a Uniform Distribution



Expected Value of Sample Mean

- Let $X_1, X_2, \dots, X_n$ represent a random sample from a population
- The **sample mean** value of these observations is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Standard Error of the Mean

- Different samples of the same size from the same population will yield different sample means
- A measure of the variability in the mean from sample to sample is given by the **Standard Error of the Mean**:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- Note that the standard error of the mean decreases as the sample size increases

If sample values are not independent (continued)

- If the sample size n is not a small fraction of the population size N , then individual sample members are not distributed independently of one another
- Thus, observations are not selected independently
- A correction is made to account for this:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n} \frac{N-n}{N-1}$$

or

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

If the Population is Normal

- If a population is **normal** with mean μ and standard deviation σ , the sampling distribution of $\bar{X}$ is **also normally distributed** with

$$\mu_{\bar{X}} = \mu$$

and

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

- If the sample size n is not large relative to the population size N , then

$$\mu_{\bar{X}} = \mu$$

and

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Z-value for Sampling Distribution of the Mean

- Z-value for the sampling distribution of :

$$Z = \frac{(\bar{X} - \mu)}{\sigma_{\bar{X}}}$$

where:

$\bar{X}$ = sample mean

μ = population mean

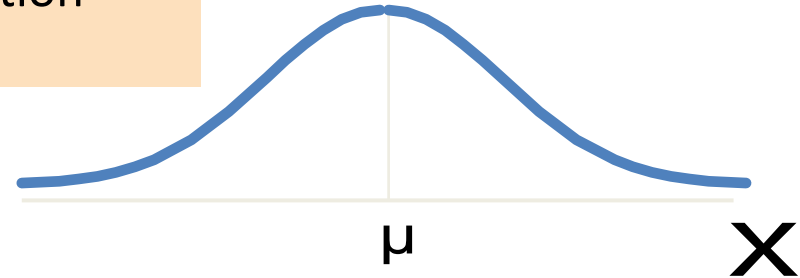
$\sigma_{\bar{X}}$ = standard error of the mean

Sampling Distribution Properties

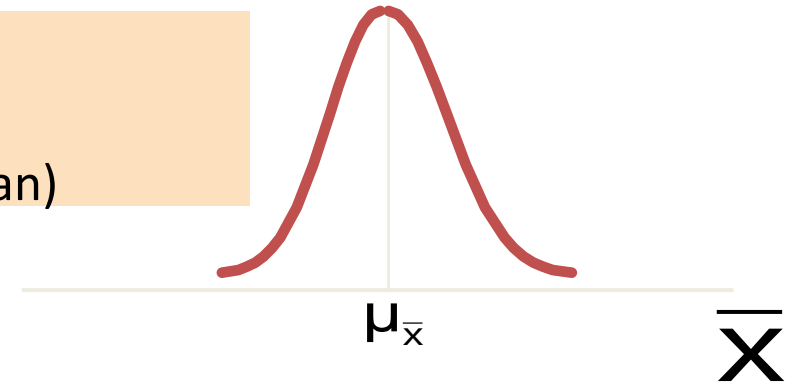
$$\mu_{\bar{X}} = \mu$$

(i.e. $\bar{X}$ is unbiased)

Normal Population
Distribution



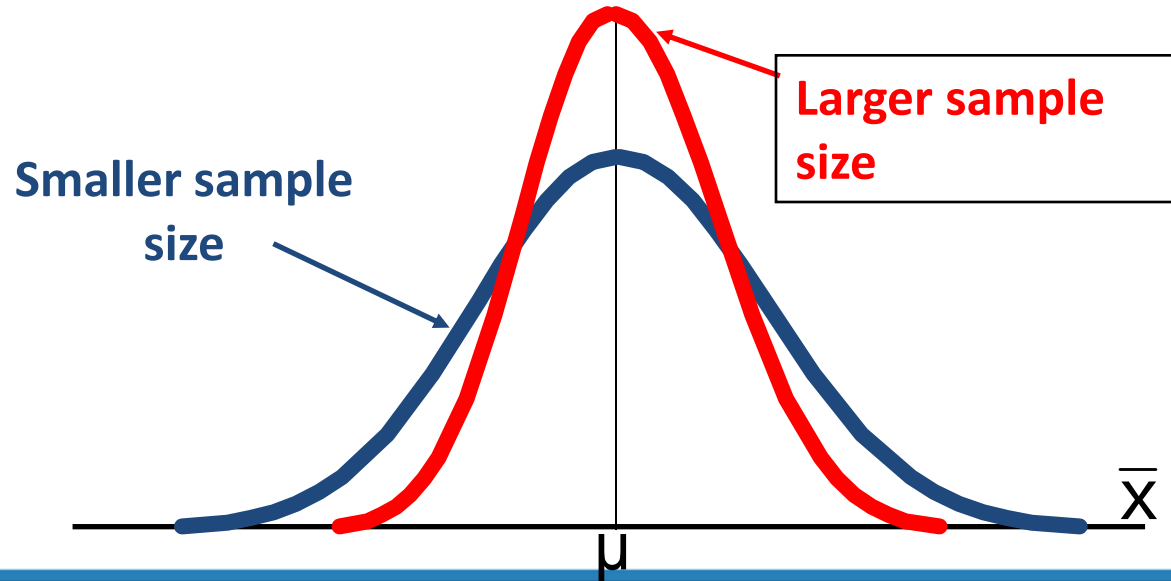
Normal Sampling
Distribution
(has the same mean)



Sampling Distribution Properties

- For sampling **with replacement**:

As n increases,
 $\sigma_{\bar{x}}$ decreases



If the Population is not Normal- Central Limit Theorem

We can apply the Central Limit Theorem:

- Even if the population is **not normal**,
- sample means from the population **will be approximately normal** as long as the sample size is large enough.

Properties of the sampling distribution:

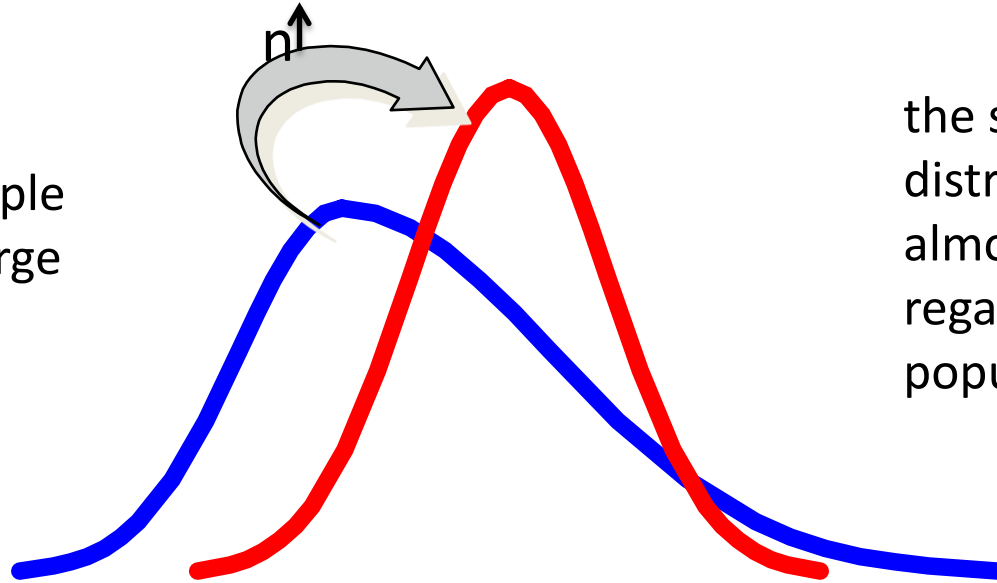
$$\mu_{\bar{x}} = \mu$$

And

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Central Limit Theorem

As the sample size gets large enough...



the sampling distribution becomes almost normal regardless of shape of population

If the Population is not Normal

(continued)

Sampling distribution properties:

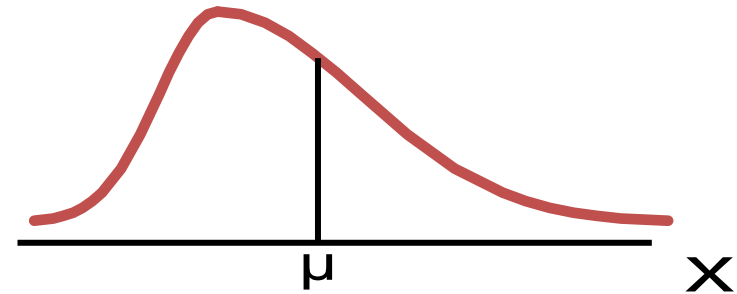
Central Tendency

$$\mu_{\bar{x}} = \mu$$

Variation

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

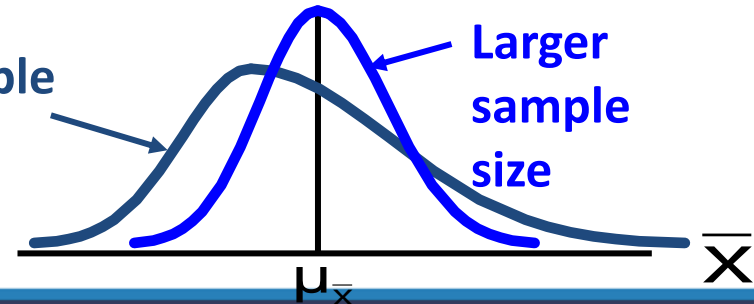
Population Distribution



Sampling Distribution (becomes normal as n increases)

Smaller sample size

Larger sample size



How Large is Large Enough?

- For most distributions, $n > 25$ will give a sampling distribution that is nearly normal
- For normal population distributions, the sampling distribution of the mean is always normally distributed

Example

- Suppose a large population has mean $\mu = 8$ and standard deviation $\sigma = 3$. Suppose a random sample of size $n = 36$ is selected.
- What is the probability that the sample mean is between 7.8 and 8.2?

Example

Solution:

- Even if the population is not normally distributed, the central limit theorem can be used ($n > 25$)
- ... so the sampling distribution of $\bar{X}$ is approximately normal
- ... with mean $\mu_{\bar{X}} = 8$
- ...and standard deviation

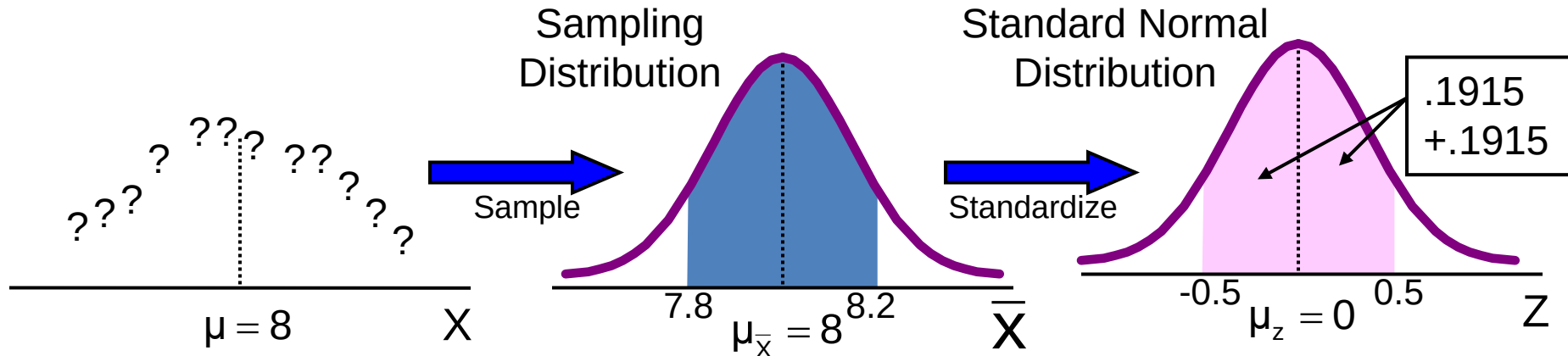
$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$$

Example

(continued)

Solution (continued)

$$\begin{aligned} P(7.8 < \mu_{\bar{X}} < 8.2) &= P\left(\frac{7.8 - 8}{\frac{3}{\sqrt{36}}} < \frac{\mu_{\bar{X}} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{8.2 - 8}{\frac{3}{\sqrt{36}}}\right) \\ &= P(-0.5 < Z < 0.5) = 0.3830 \end{aligned}$$





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Distribution of Sample Mean, proportion, and variance

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Acceptance Intervals

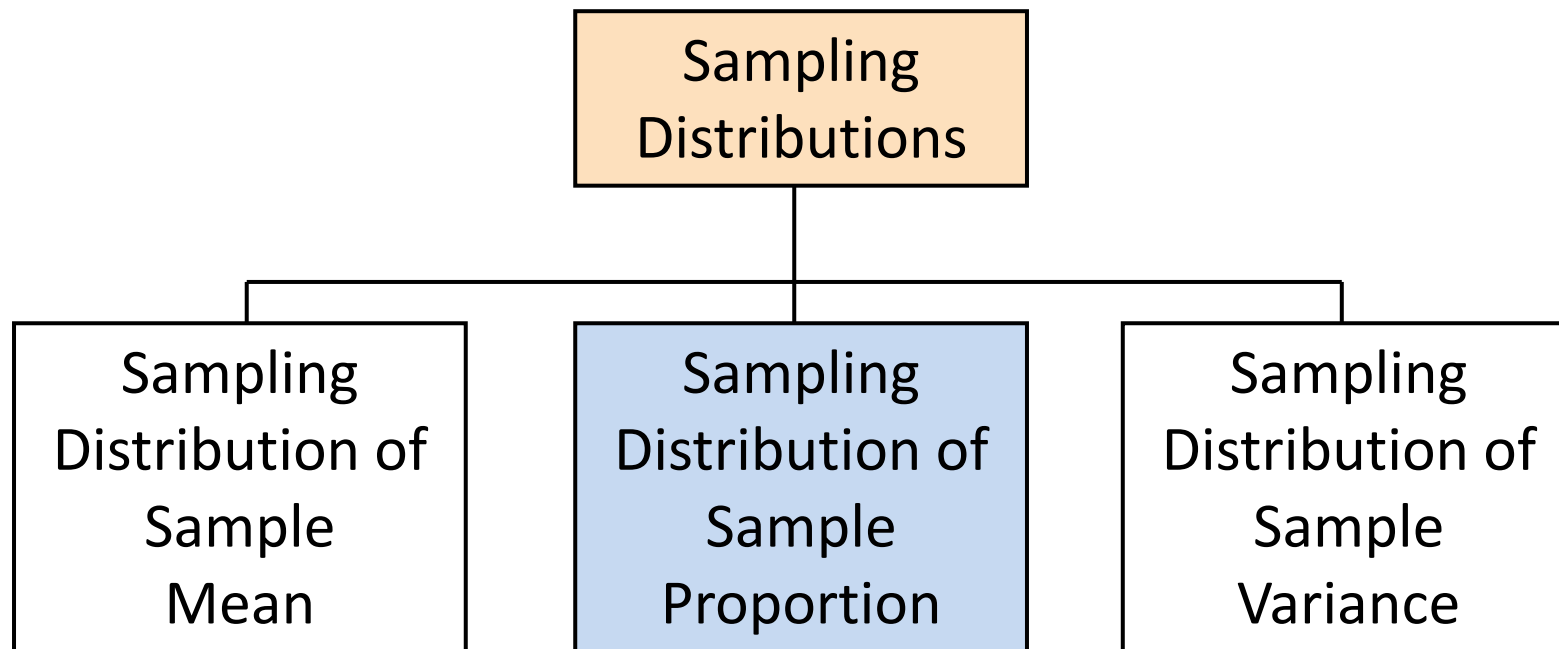
Goal: determine a range within which sample means are likely to occur, given a population mean and variance

- By the Central Limit Theorem, we know that the distribution of $\bar{X}$ is approximately normal if n is large enough, with mean μ and standard deviation
- Let $z_{\alpha/2}$ be the z -value that leaves area $\alpha/2$ in the upper tail of the normal distribution (i.e., the interval $-\bar{z}_{\alpha/2}$ to $\bar{z}_{\alpha/2}$ encloses probability $1 - \alpha$)
- Then

$$\mu \pm z_{\alpha/2} \sigma_{\bar{X}}$$

is the interval that includes $\bar{X}$ with probability $1 - \alpha$

Sampling Distributions of Sample Proportions



Sampling Distributions of Sample Proportions

P = the proportion of the population having some characteristic

- **Sample proportion** ($\hat{p}$) provides an estimate of P :

$$\hat{p} = \frac{X}{n} = \frac{\text{number of items in the sample having the characteristic of interest}}{\text{sample size}}$$

- $0 \leq \hat{p} \leq 1$
- $\hat{p}$ has a binomial distribution, but can be approximated by a normal distribution when $nP(1 - P) > 5$

Sampling Distribution of $\hat{p}$

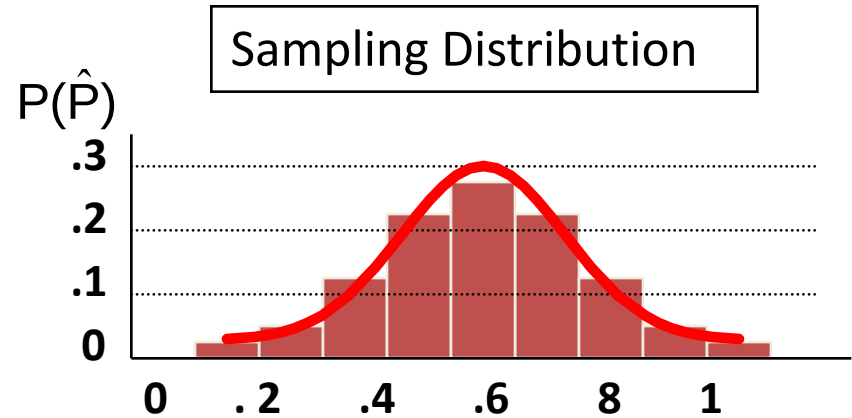
- Normal approximation:

Properties: $E(\hat{p}) = P$

(where P = population proportion)

And

$$\sigma_{\hat{p}}^2 = \text{Var}\left(\frac{X}{n}\right) = \frac{P(1-P)}{n}$$





Z-Value for Proportions

Standardize $\hat{p}$ to a Z value with the formula:

$$Z = \frac{\hat{p} - P}{\sigma_{\hat{p}}} = \frac{\hat{p} - P}{\sqrt{\frac{P(1-P)}{n}}}$$

Example

- If the true proportion of voters who support Proposition A is $P = .4$, what is the probability that a sample of size 200 yields a sample proportion between .40 and .45?
- i.e.:

if $P = .4$ and $n = 200$, what is
 $P(.40 \leq \hat{p} \leq .45)$?

Example

(continued)

- if $P = .4$ and $n = 200$, what is $P(.40 \leq \hat{p} \leq .45)$?

Find: $\sigma_{\hat{p}}$

$$\sigma_{\hat{p}} = \sqrt{\frac{P(1-P)}{n}} = \sqrt{\frac{.4(1-.4)}{200}} = .03464$$

Convert to
standard
normal:

$$\begin{aligned} P(.40 \leq \hat{p} \leq .45) &= P\left(\frac{.40 - .40}{.03464} \leq Z \leq \frac{.45 - .40}{.03464}\right) \\ &= P(0 \leq Z \leq 1.44) \end{aligned}$$

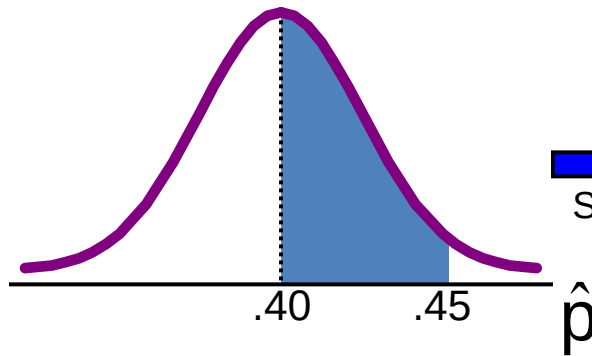
Example

(continued)

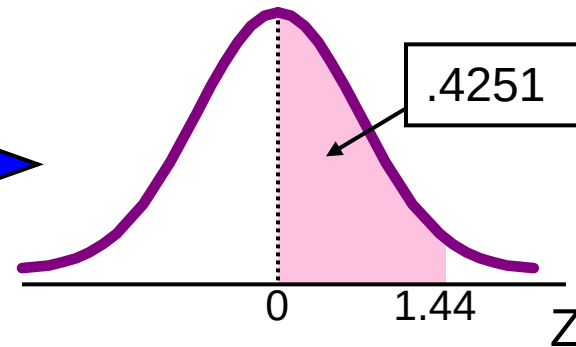
if $P = .4$ and $n = 200$, what is $P(.40 \leq \hat{p} \leq .45)$?

Use standard normal table: $P(0 \leq Z \leq 1.44) = .4251$

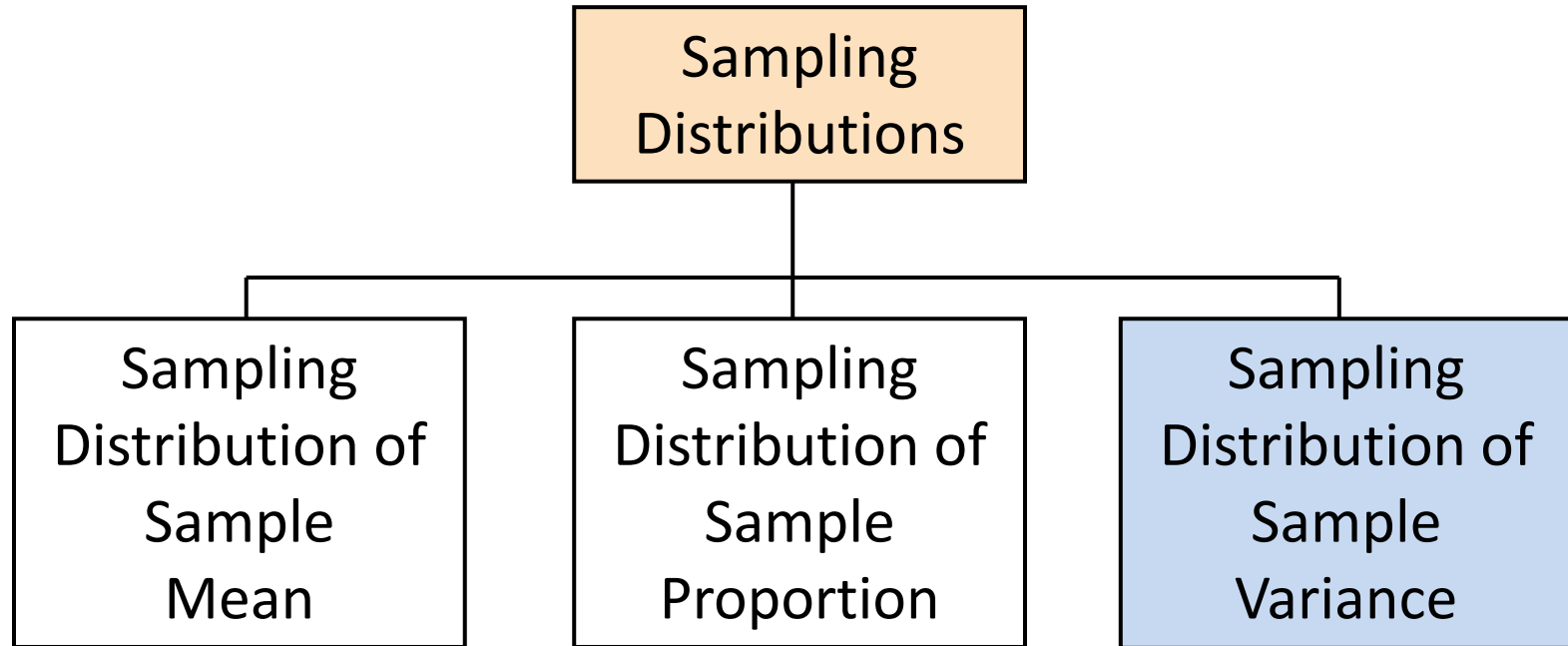
Sampling Distribution



Standardized
Normal Distribution



Sampling Distributions of Sample Variance



Sample Variance

- Let $x_1, x_2, \dots, x_n$ be a random sample from a population. The **sample variance** is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- the square root of the sample variance is called the **sample standard deviation**
- the sample variance is different for different random samples from the same population

Sampling Distribution of Sample Variances

- The sampling distribution of s^2 has mean σ^2

$$E(s^2) = \sigma^2$$

- If the population distribution is normal then

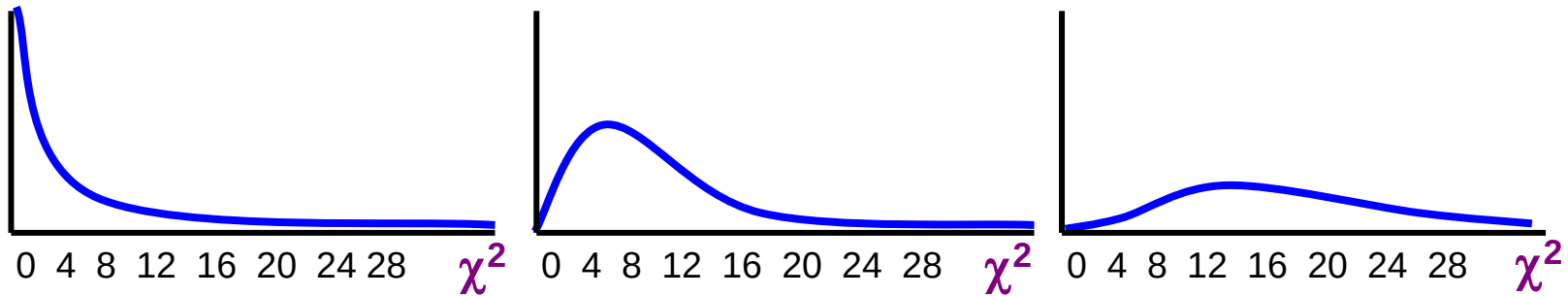
$$\frac{(n-1)s^2}{\sigma^2}$$

has a χ^2 distribution with $n - 1$ degrees of freedom



The Chi-square Distribution

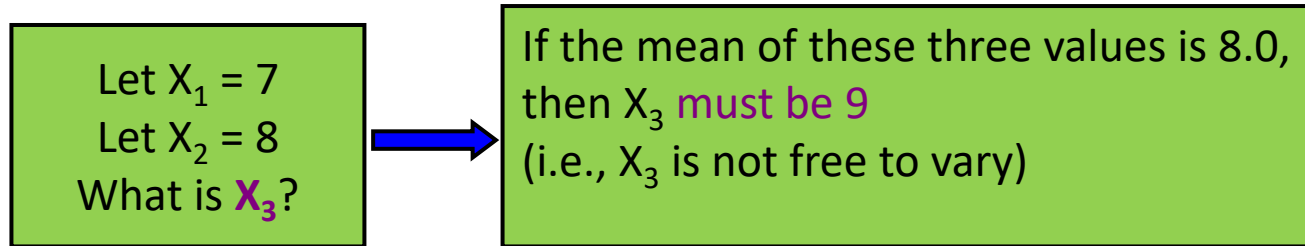
- The **chi-square distribution** is a family of distributions, depending on degrees of freedom: **d.f. = $n - 1$**



Degrees of Freedom (df)

Idea: Number of observations that are free to vary after sample mean has been calculated

Example: Suppose the mean of 3 numbers is 8.0



Here, $n = 3$, so degrees of freedom $= n - 1 = 3 - 1 = 2$

(2 values can be any numbers, but the third is not free to vary for a given mean)

Chi-square Example

- A commercial freezer must hold a selected temperature with little variation. Specifications call for a standard deviation of no more than 4 degrees (a variance of 16 degrees²).
- A sample of 14 freezers is to be tested
- What is the upper limit (K) for the sample variance such that the probability of exceeding this limit, given that the population standard deviation is 4, is less than 0.05?

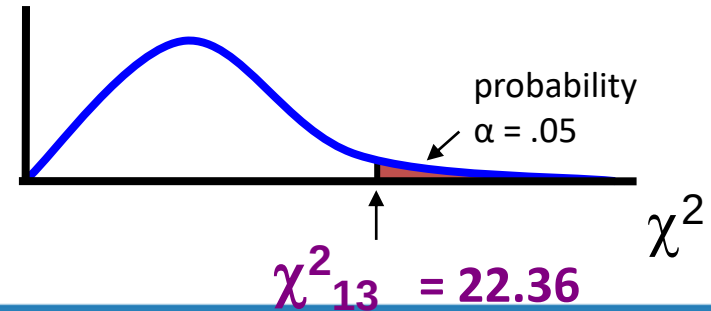
Finding the Chi-square Value

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

Is chi-square distributed with $(n - 1) = 13$ degrees of freedom

- Use the the chi-square distribution with area 0.05 in the upper tail:

$$\chi^2_{13} = 22.36 \quad (\alpha = .05 \text{ and } 14 - 1 = 13 \text{ d.f.})$$



Chi-square Example

(continued)

$$\chi^2_{13} = 22.36 \quad (\alpha = .05 \text{ and } 14 - 1 = 13 \text{ d.f.})$$

So:

$$P(s^2 > K) = P\left(\frac{(n-1)s^2}{16} > \chi^2_{13}\right) = 0.05$$

$$\text{or} \quad \frac{(n-1)K}{16} = 22.36$$

$$\text{so} \quad K = \frac{(22.36)(16)}{(14-1)} = 27.52$$

(where $n = 14$)

If s^2 from the sample of size $n = 14$ is greater than 27.52, there is strong evidence to suggest the population variance exceeds 16.

Summary

- Introduced sampling distributions
- Described the sampling distribution of sample means
 - For normal populations
 - Using the Central Limit Theorem
- Described the sampling distribution of sample proportions
- Introduced the chi-square distribution
- Examined sampling distributions for sample variances
- Calculated probabilities using sampling distributions

Thank You





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Confidence Interval Estimation: Single Population

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Goals

After completing this lecture, you should be able to:

- Distinguish between a point estimate and a confidence interval estimate
- Construct and interpret a confidence interval estimate for a single population mean using both the Z and t distributions
- Form and interpret a confidence interval estimate for a single population proportion
- Create confidence interval estimates for the variance of a normal population

Confidence Intervals

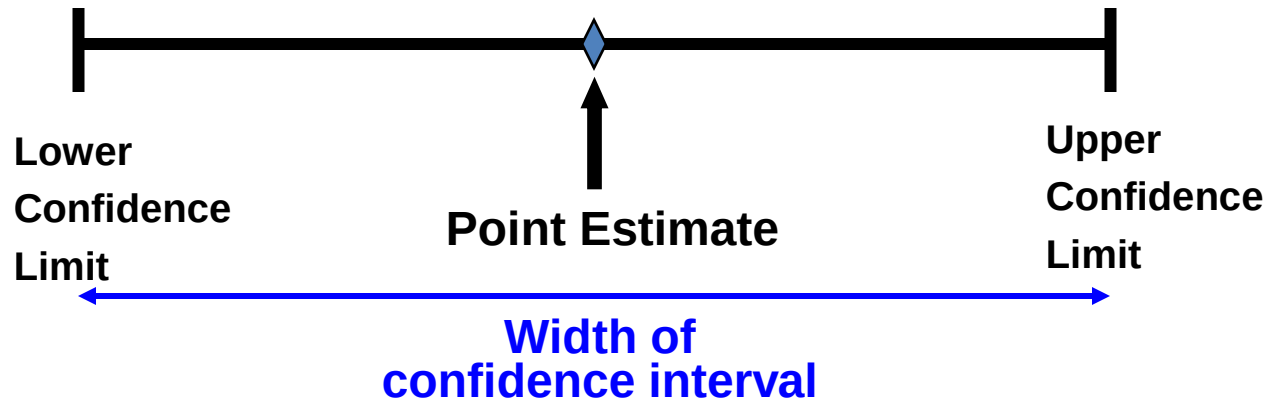
- Confidence Intervals for the Population Mean, μ
 - when Population Variance σ^2 is Known
 - when Population Variance σ^2 is Unknown
- Confidence Intervals for the Population Proportion, $\hat{p}$ (large samples)
- Confidence interval estimates for the variance of a normal population

Definitions

- An **estimator** of a population parameter is
 - a random variable that depends on sample information . . .
 - whose value provides an approximation to this unknown parameter
- A specific value of that random variable is called an **estimate**

Point and Interval Estimates

- A **point estimate** is a single number,
- a **confidence interval** provides additional information about variability



Point Estimates

We can estimate a Population Parameter ...		with a Sample Statistic (a Point Estimate)
Mean	μ	$\bar{x}$
Proportion	P	$\hat{p}$

Unbiasedness

- A point estimator $\hat{\theta}$ is said to be an **unbiased estimator** of the parameter θ if the expected value, or mean, of the sampling distribution of $\hat{\theta}$ is θ ,

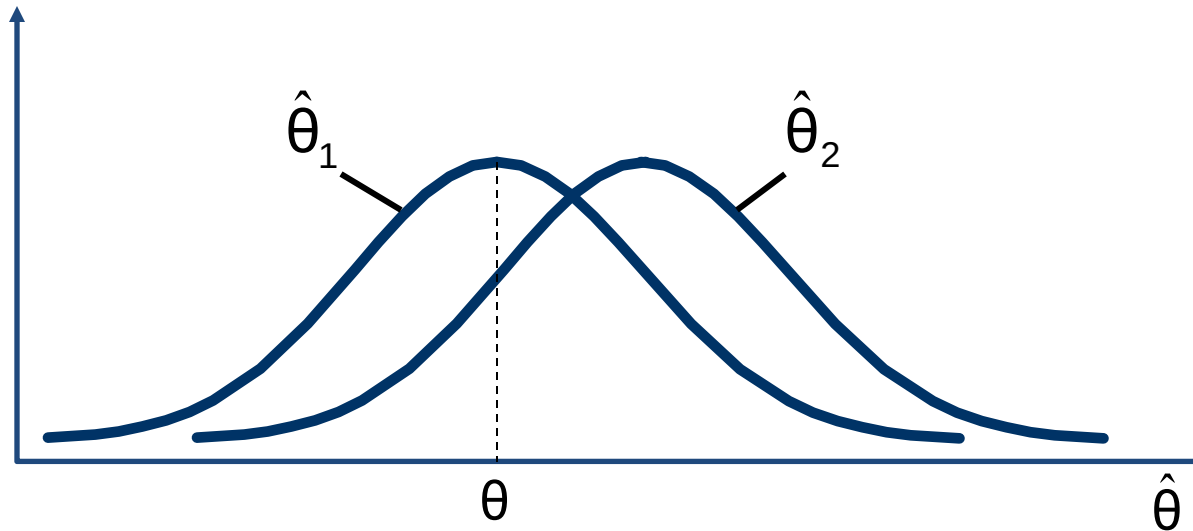
$$E(\hat{\theta}) = \theta$$

- Examples:
 - The sample mean $\bar{x}$ is an unbiased estimator of μ
 - The sample variance s^2 is an unbiased estimator of σ^2
 - The sample proportion $\hat{p}$ is an unbiased estimator of P

Unbiasedness

(continued)

- $\hat{\theta}_1$ is an unbiased estimator, $\hat{\theta}_2$ is biased:



Bias

- Let $\hat{\theta}$ be an estimator of θ
- The **bias** in $\hat{\theta}$ is defined as the difference between its mean and θ

$$\text{Bias}(\hat{\theta}) = E(\hat{\theta}) - \theta$$

- The bias of an unbiased estimator is 0

Most Efficient Estimator

- Suppose there are several unbiased estimators of θ
- The **most efficient estimator** or the **minimum variance unbiased estimator** of θ is the unbiased estimator with the **smallest variance**
- Let $\hat{\theta}_1$ and $\hat{\theta}_2$ be two unbiased estimators of θ , based on the same number of sample observations. Then,
 - $\hat{\theta}_1$ is said to be more efficient than $\hat{\theta}_2$ if $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$
 - The relative efficiency of $\hat{\theta}_1$ with respect to $\hat{\theta}_2$ is the ratio of their variances:

$$\text{Relative Efficiency} = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

Confidence Intervals

- How much uncertainty is associated with a point estimate of a population parameter?
- An **interval estimate** provides more information about a population characteristic than does a **point estimate**
- Such interval estimates are called confidence intervals

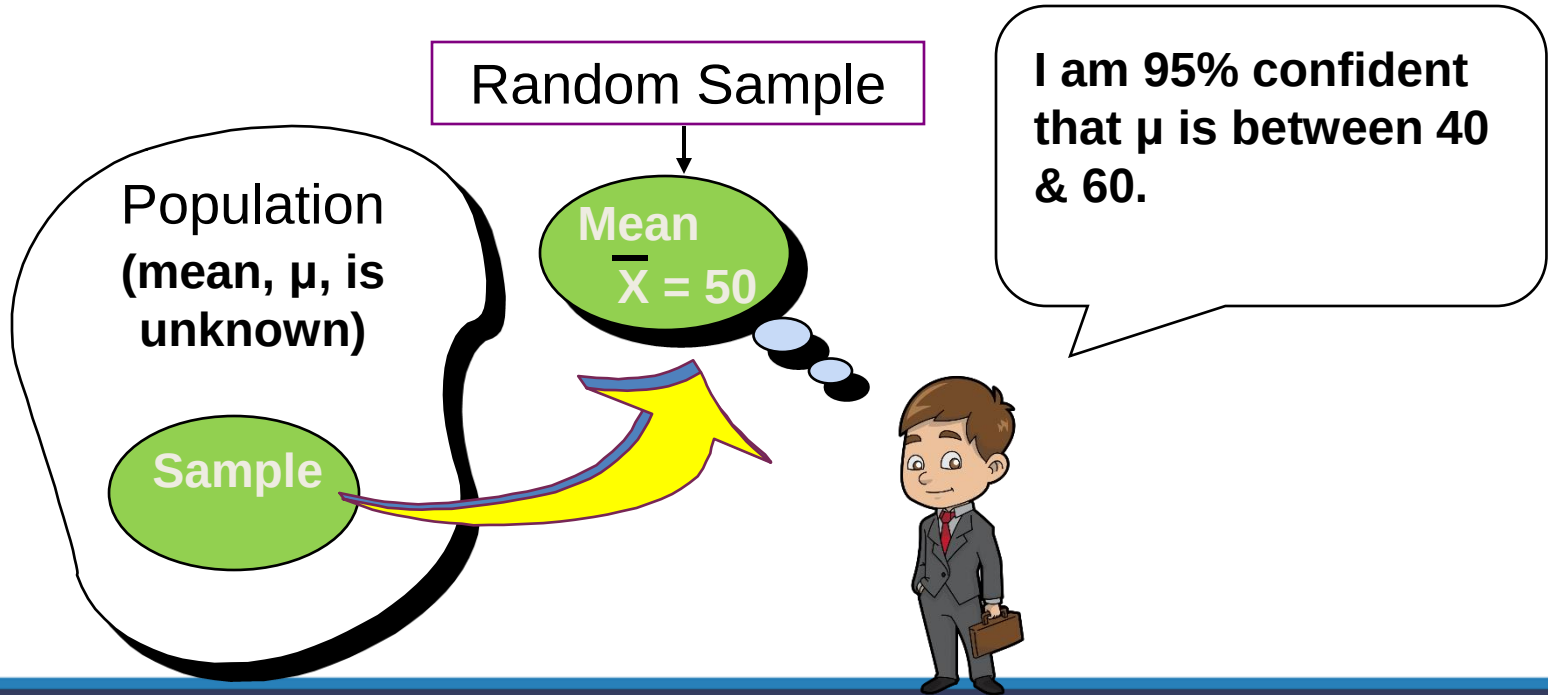
Confidence Interval Estimate

- An interval gives a **range** of values:
 - Takes into consideration variation in sample statistics from sample to sample
 - Based on observation from 1 sample
 - Gives information about closeness to unknown population parameters
 - Stated in terms of level of confidence
 - Can never be 100% confident

Confidence Interval and Confidence Level

- If $P(a < \theta < b) = 1 - \alpha$ then the interval from a to b is called a $100(1 - \alpha)\%$ confidence interval of θ .
- The quantity $(1 - \alpha)$ is called the confidence level of the interval (α between 0 and 1)
 - In repeated samples of the population, the true value of the parameter θ would be contained in $100(1 - \alpha)\%$ of intervals calculated this way.
 - The confidence interval calculated in this manner is written as $a < \theta < b$ with $100(1 - \alpha)\%$ confidence

Estimation Process



Confidence Level, $(1-\alpha)$

(continued)

- Suppose confidence level = 95%
- Also written $(1 - \alpha) = 0.95$
- A relative frequency interpretation:
 - From repeated samples, 95% of all the confidence intervals that can be constructed will contain the unknown true parameter
- A specific interval either will contain or will not contain the true parameter



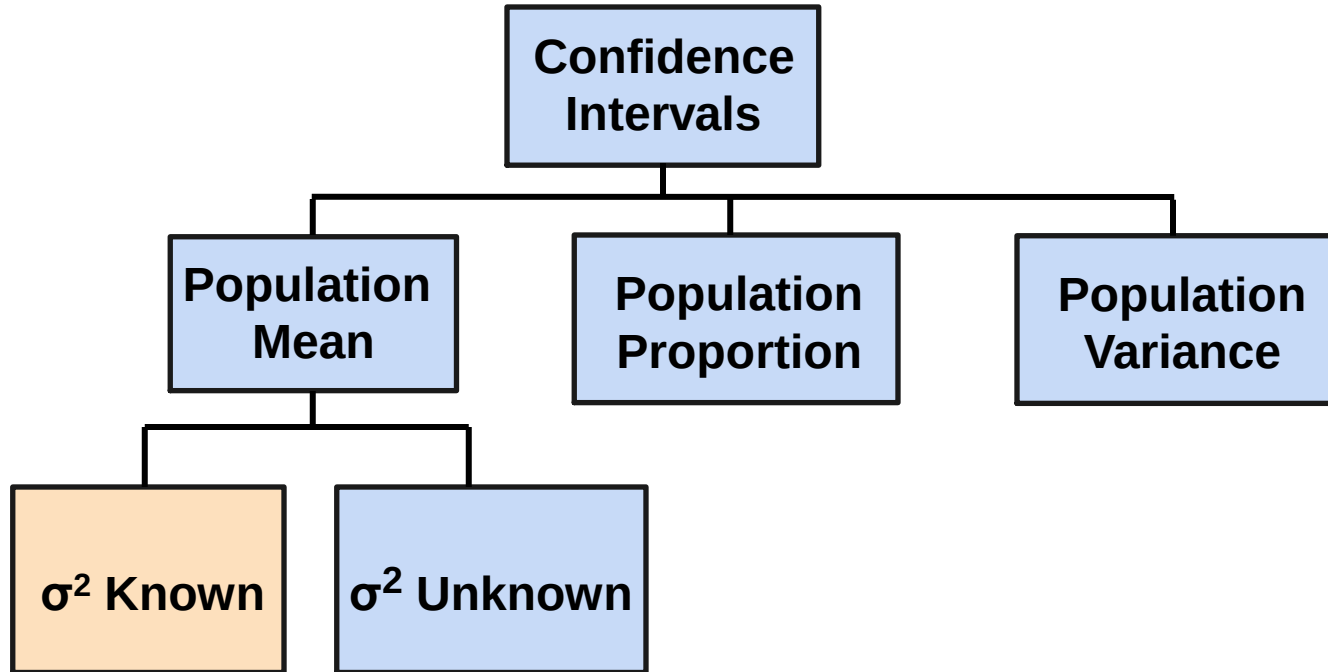
General Formula

- The general formula for all confidence intervals is:

$$\text{Point Estimate} \pm (\text{Reliability Factor})(\text{Standard Error})$$

- The value of the reliability factor depends on the desired level of confidence

Confidence Intervals



Confidence Interval for μ (σ^2 Known)

- Assumptions
 - Population variance σ^2 is known
 - Population is normally distributed
 - If population is not normal, use large sample
- Confidence interval estimate:

$$\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

(where $z_{\alpha/2}$ is the normal distribution value for a probability of $\alpha/2$ in each tail)

Margin of Error

- The confidence interval,

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- Can also be written as $\bar{x} \pm ME$
where ME is called the **margin of error**

$$ME = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

Reducing the Margin of Error

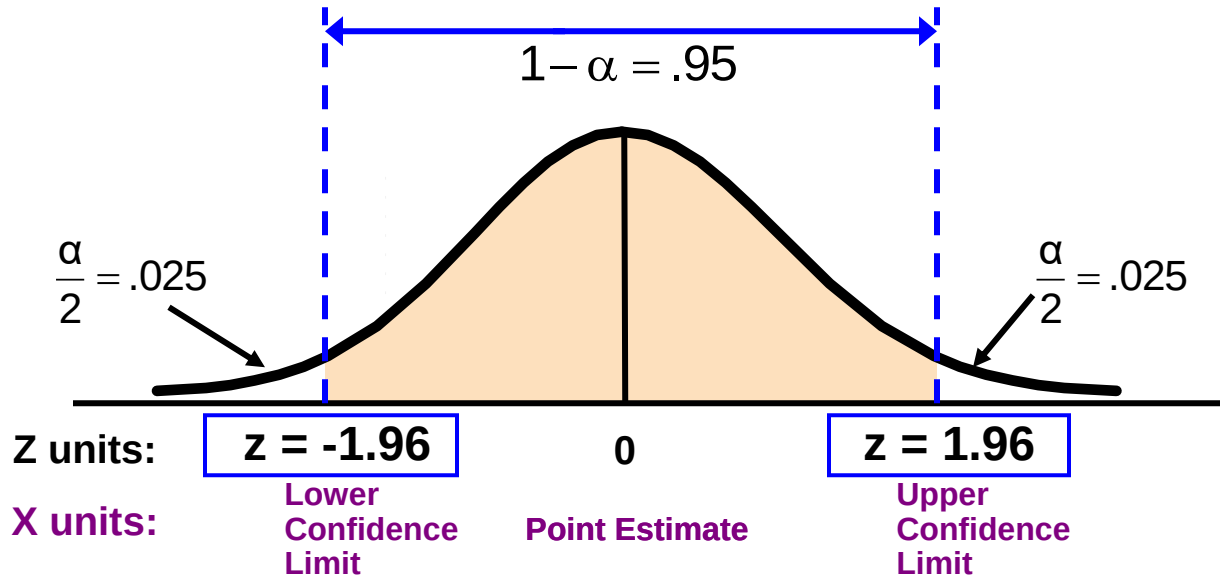
$$ME = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The margin of error can be reduced if

- the population standard deviation can be reduced ($\sigma \downarrow$)
- The sample size is increased ($n \uparrow$)
- The confidence level is decreased, $(1 - \alpha) \downarrow$

Finding the Reliability Factor, $z_{\alpha/2}$

- Consider a 95% confidence interval:



- Find $z_{.025} = \pm 1.96$ from the standard normal distribution table

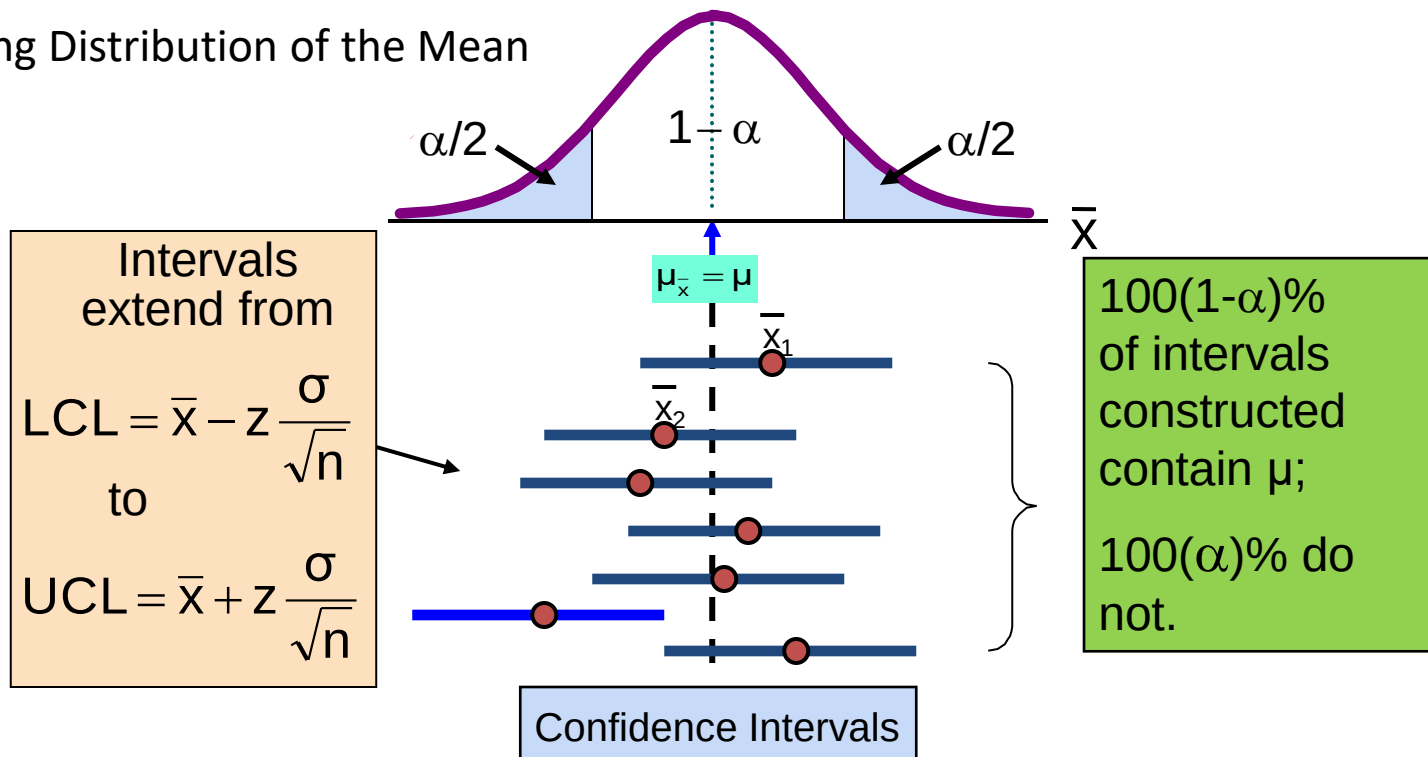
Common Levels of Confidence

- Commonly used confidence levels are 90%, 95%, and 99%

<i>Confidence Level</i>	<i>Confidence Coefficient, $1 - \alpha$</i>	<i>$Z_{\alpha/2}$ value</i>
80%	.80	1.28
90%	.90	1.645
95%	.95	1.96
98%	.98	2.33
99%	.99	2.58
99.8%	.998	3.08
99.9%	.999	3.27

Intervals and Level of Confidence

Sampling Distribution of the Mean



Example

- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms.
- Determine a 95% confidence interval for the true mean resistance of the population.

Example

(continued)

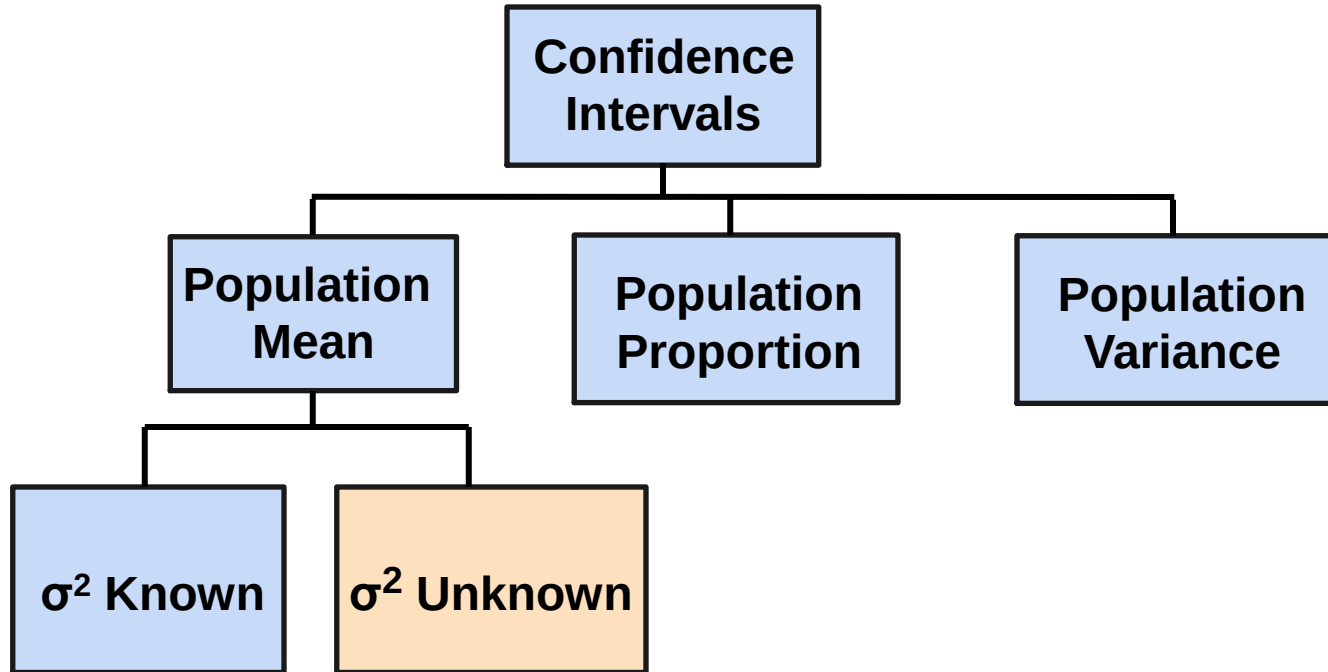
- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is .35 ohms.

- **Solution:**
$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$
$$= 2.20 \pm 1.96 (.35/\sqrt{11})$$
$$= 2.20 \pm .2068$$
$$1.9932 < \mu < 2.4068$$

Interpretation

- We are 95% confident that the true mean resistance is between 1.9932 and 2.4068 ohms
- Although the true mean may or may not be in this interval, 95% of intervals formed in this manner will contain the true mean

Confidence Intervals





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Confidence Interval Estimation: Single Population-II

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Student's t Distribution

- Consider a random sample of n observations
 - with mean $\bar{x}$ and standard deviation s
 - from a normally distributed population with mean μ

- Then the variable

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

follows the **Student's t distribution** with $(n - 1)$ degrees of freedom

Confidence Interval for μ (σ^2 Unknown)

- If the population standard deviation σ is unknown, we can substitute the sample standard deviation, s
- This introduces extra uncertainty, since s is variable from sample to sample
- So we use the t distribution instead of the normal distribution

Confidence Interval for μ (σ Unknown)

(continued)

- Assumptions
 - Population standard deviation is unknown
 - Population is normally distributed
 - If population is not normal, use large sample
- Use Student's t Distribution
- Confidence Interval Estimate:

$$\bar{x} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}$$

where $t_{n-1, \alpha/2}$ is the critical value of the t distribution with $n-1$ d.f. and an area of $\alpha/2$ in each tail

Margin of Error

- The confidence interval,

$$\bar{x} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

- Can also be written as

$$\bar{x} \pm ME$$

where ME is called the margin of error:

$$ME = t_{n-1, \alpha/2} \frac{\sigma}{\sqrt{n}}$$

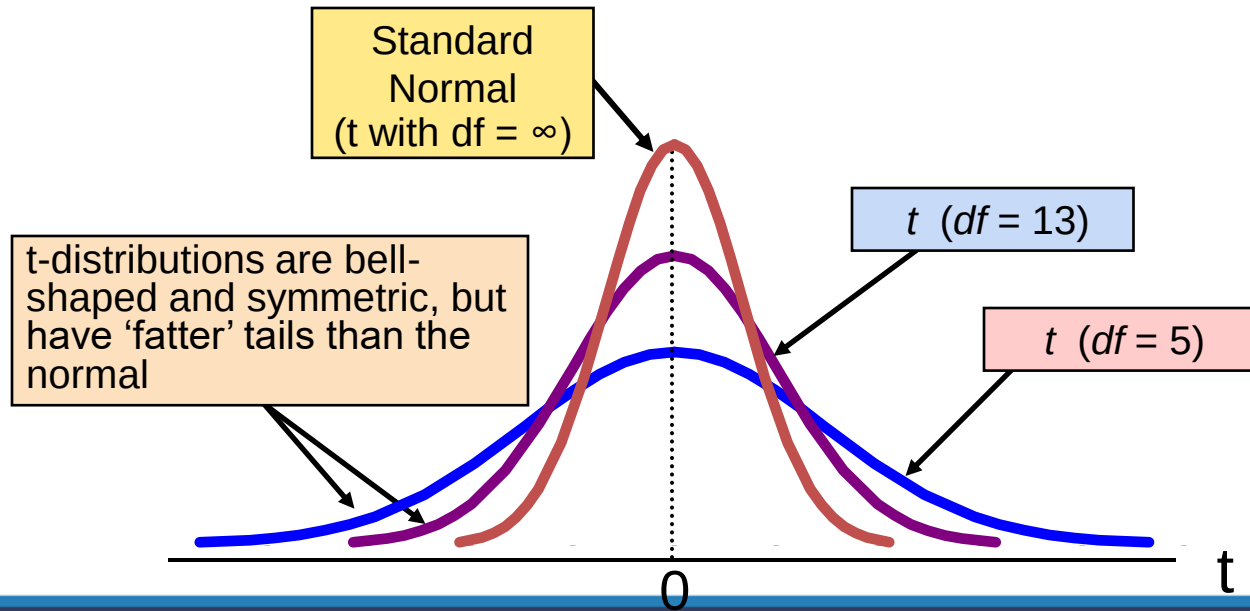
Student's t Distribution

- The t is a family of distributions
- The t value depends on degrees of freedom (d.f.)
 - Number of observations that are free to vary after sample mean has been calculated

$$\text{d.f.} = n - 1$$

Student's t Distribution

Note: $t \rightarrow Z$ as n increases

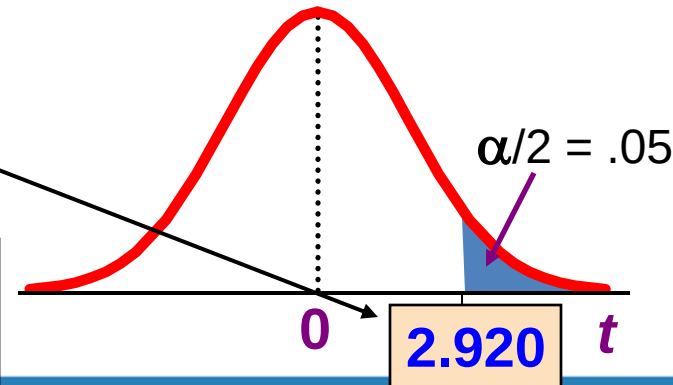


Student's t Table

Upper Tail Area			
df	.10	.05	.025
1	3.078	6.314	12.706
2	1.886	2.920	4.303
3	1.638	2.353	3.182

Let: $n = 3$
 $df = n - 1 = 2$
 $\alpha = .10$
 $\alpha/2 = .05$

The body of the table contains t values, not probabilities



t distribution values

With comparison to the Z value

Confidence Level	t (10 d.f.)	t (20 d.f.)	t (30 d.f.)	Z
.80	1.372	1.325	1.310	1.282
.90	1.812	1.725	1.697	1.645
.95	2.228	2.086	2.042	1.960
.99	3.169	2.845	2.750	2.576

Note: $t \rightarrow Z$ as n increases

Example

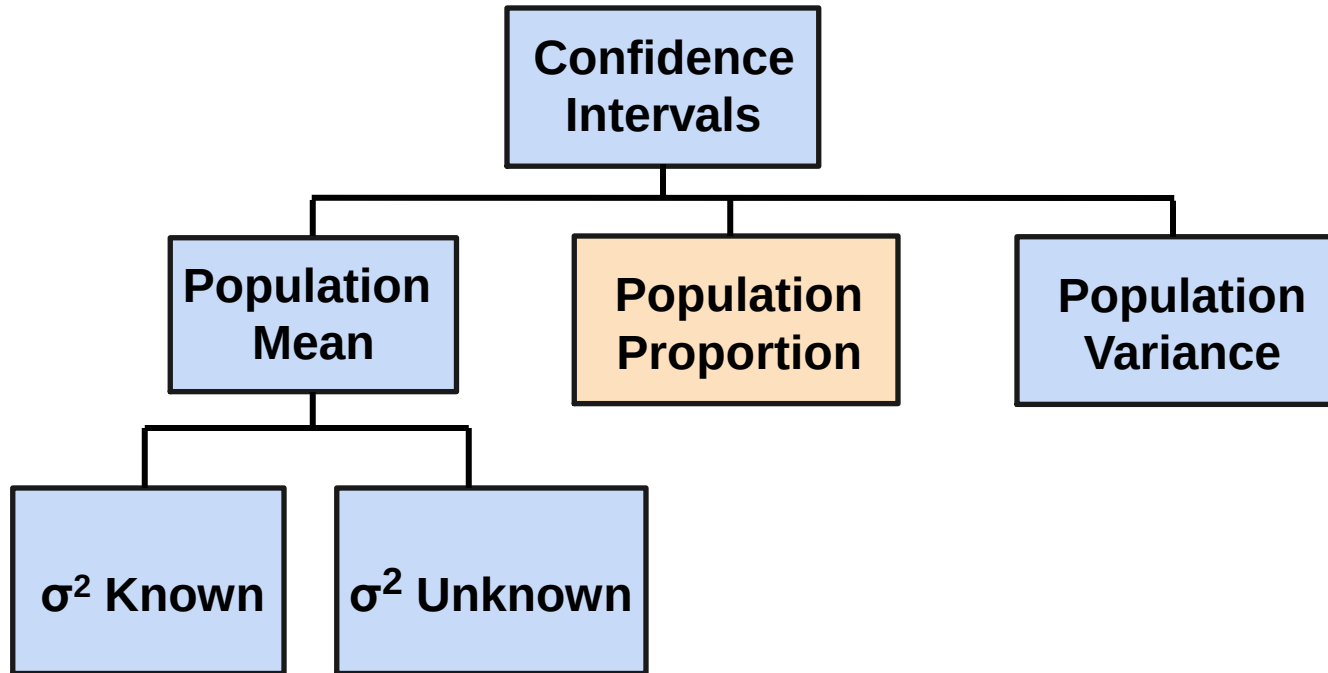
A random sample of $n = 25$ has $\bar{x} = 50$ and $s = 8$. Form a 95% confidence interval for μ

$$- \text{d.f.} = n - 1 = 24, \text{ so} \quad t_{n-1, \alpha/2} = t_{24, .025} = 2.0639$$

The confidence interval is

$$\begin{aligned} \bar{x} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \\ 50 - (2.0639) \frac{8}{\sqrt{25}} < \mu < 50 + (2.0639) \frac{8}{\sqrt{25}} \\ 46.698 < \mu < 53.302 \end{aligned}$$

Confidence Intervals



Confidence Intervals for the Population Proportion

- An interval estimate for the population proportion (P) can be calculated by adding an allowance for uncertainty to the sample proportion ($\hat{p}$)

Confidence Intervals for the Population Proportion, p

(continued)

- Recall that the distribution of the sample proportion is approximately normal if the sample size is large, with standard deviation

$$\sigma_P = \sqrt{\frac{P(1-P)}{n}}$$

- We will estimate this with sample data:

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Confidence Interval Endpoints

- Upper and lower confidence limits for the population proportion are calculated with the formula

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < P < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- where
 - $z_{\alpha/2}$ is the standard normal value for the level of confidence desired
 - $\hat{p}$ is the sample proportion
 - n is the sample size
 - $nP(1-P) > 5$

Example

- A random sample of 100 people shows that 25 are left-handed.
- Form a 95% confidence interval for the true proportion of left-handers

Example

(continued)

- A random sample of 100 people shows that 25 are left-handed. Form a 95% confidence interval for the true proportion of left-handers.

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < P < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

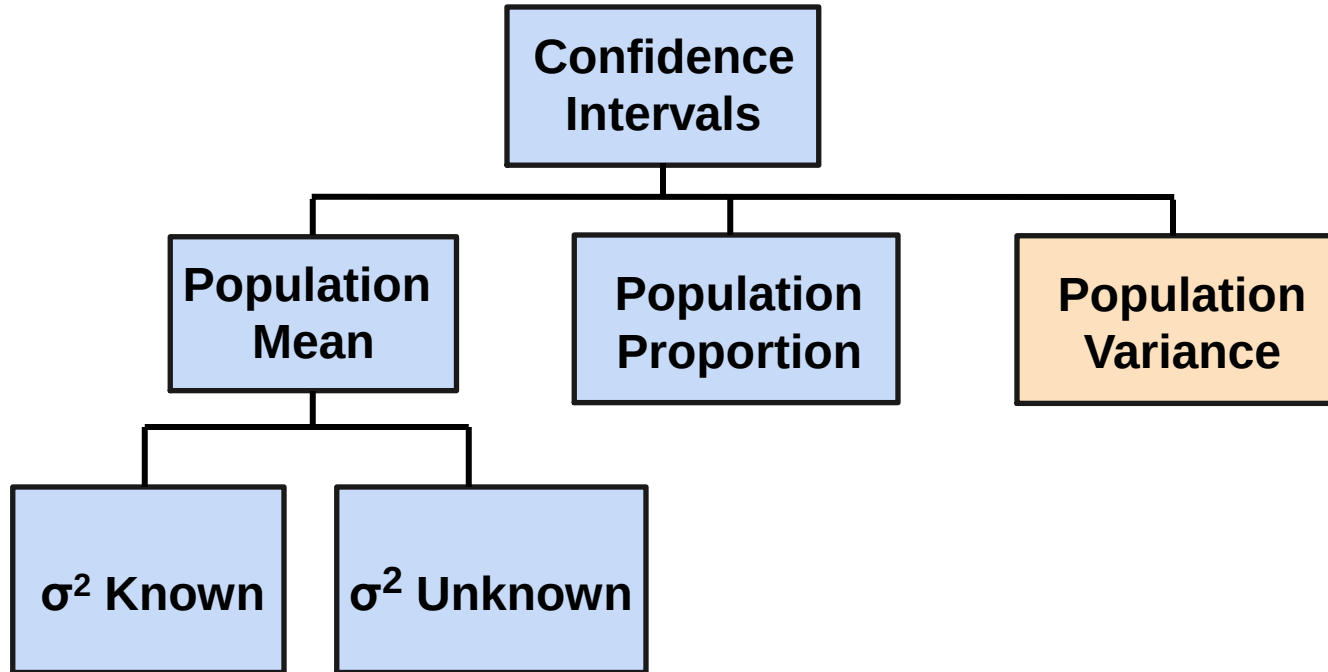
$$\frac{25}{100} - 1.96 \sqrt{\frac{.25(.75)}{100}} < P < \frac{25}{100} + 1.96 \sqrt{\frac{.25(.75)}{100}}$$

$$0.1651 < P < 0.3349$$

Interpretation

- We are 95% confident that the true percentage of left-handers in the population is between
16.51% and 33.49%.
- Although the interval from 0.1651 to 0.3349 may or may not contain the true proportion, 95% of intervals formed from samples of size 100 in this manner will contain the true proportion.

Confidence Intervals



Confidence Intervals for the Population Variance

- **Goal:** Form a confidence interval for the population variance, σ^2
 - The confidence interval is based on the sample variance, s^2
 - Assumed: the population is normally distributed

Confidence Intervals for the Population Variance

(continued)

The random variable

$$\chi_{n-1}^2 = \frac{(n-1)s^2}{\sigma^2}$$

follows a chi-square distribution with $(n - 1)$ degrees of freedom

Confidence Intervals for the Population Variance

The $(1 - \alpha)\%$ confidence interval for the population variance is

$$\frac{(n-1)s^2}{\chi_{n-1, \alpha/2}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{n-1, 1-\alpha/2}^2}$$

Example

You are testing the speed of a batch of computer processors. You collect the following data (in Mhz):

Sample size 17

Sample mean 3004

Sample std dev 74

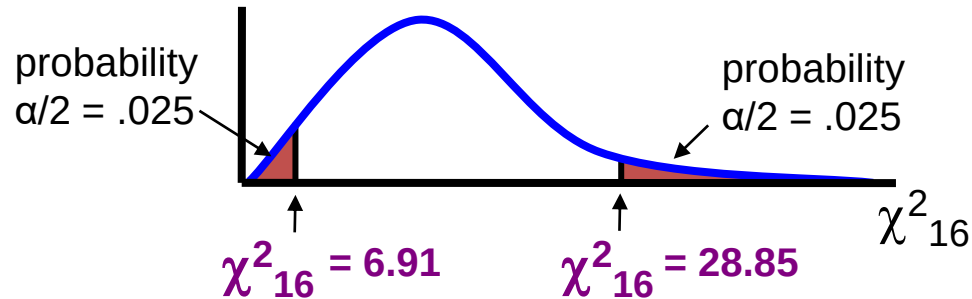
Assume the population is normal. Determine the 95% confidence interval for σ_x^2

Finding the Chi-square Values

- $n = 17$ so the chi-square distribution has $(n - 1) = 16$ degrees of freedom
- $\alpha = 0.05$, so use the the chi-square values with area 0.025 in each tail:

$$\chi^2_{n-1, \alpha/2} = \chi^2_{16, 0.025} = 28.85$$

$$\chi^2_{n-1, 1-\alpha/2} = \chi^2_{16, 0.975} = 6.91$$



Calculating the Confidence Limits

- The 95% confidence interval is

$$\frac{(n-1)s^2}{\chi_{n-1, \alpha/2}^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_{n-1, 1-\alpha/2}^2}$$

$$\frac{(17-1)(74)^2}{28.85} < \sigma^2 < \frac{(17-1)(74)^2}{6.91}$$

$$3037 < \sigma^2 < 12683$$

Converting to standard deviation, we are 95% confident that the population standard deviation of CPU speed is between 55.1 and 112.6 Mhz

Finite Populations

- If the sample size is more than 5% of the population size (and sampling is without replacement) then a **finite population correction factor** must be used when calculating the standard error

Finite Population Correction Factor

- Suppose sampling is **without replacement** and the sample size is large relative to the population size
- Assume the population size is large enough to apply the central limit theorem
- Apply the **finite population correction factor** when estimating the population variance

$$\text{finite population correction factor} = \frac{N-n}{N-1}$$

Estimating the Population Mean

- Let a simple random sample of size n be taken from a population of N members with mean μ
- The sample mean is an **unbiased estimator** of the population mean μ
- The **point estimate** is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Finite Populations: Mean

- If the sample size is more than 5% of the population size, an unbiased estimator for the variance of the sample mean is

$$\hat{\sigma}_{\bar{x}}^2 = \frac{s^2}{n} \left(\frac{N-n}{N-1} \right)$$

- So the 100(1- α)% confidence interval for the population mean is

$$\bar{X} - t_{n-1, \alpha/2} \hat{\sigma}_{\bar{x}} < \mu < \bar{X} + t_{n-1, \alpha/2} \hat{\sigma}_{\bar{x}}$$

Estimating the Population Proportion

- Let the true population proportion be P
- Let $\hat{p}$ be the sample proportion from n observations from a simple random sample
- The sample proportion, $\hat{p}$, is an unbiased estimator of the population proportion, P

Finite Populations: Proportion

- If the sample size is more than 5% of the population size, an unbiased estimator for the variance of the population proportion is

$$\hat{\sigma}_{\hat{p}}^2 = \frac{\hat{p}(1-\hat{p})}{n} \left(\frac{N-n}{N-1} \right)$$

- So the 100(1- α)% confidence interval for the population proportion is

$$\hat{p} - z_{\alpha/2} \hat{\sigma}_{\hat{p}} < P < \hat{p} + z_{\alpha/2} \hat{\sigma}_{\hat{p}}$$

Lecture Summary

- Introduced the concept of confidence intervals
- Discussed point estimates
- Developed confidence interval estimates
- Created confidence interval estimates for the mean (σ^2 known)
- Introduced the Student's t distribution
- Determined confidence interval estimates for the mean (σ^2 unknown)

Lecture Summary

(continued)

- Created confidence interval estimates for the proportion
- Created confidence interval estimates for the variance of a normal population
- Applied the finite population correction factor to form confidence intervals when the sample size is not small relative to the population size

Summary

- Introduced sampling distributions
- Described the sampling distribution of sample means
 - For normal populations
 - Using the Central Limit Theorem
- Described the sampling distribution of sample proportions
- Introduced the chi-square distribution
- Examined sampling distributions for sample variances
- Calculated probabilities using sampling distributions

Thank You

